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DCInstrumentationDefTopic



DCTime Theory of Operation

DCTime Theory of Operation

A DC30 or DC75 DC Time Measurement Unit (DCTime) can perform time measurements, such as rise and fall time and measurements of period, frequency, and pulse width. Measurements are performed by detecting events generated by the DUT or other instruments. Each DC30 or DC75 board includes two DCTimes.

Time measurements are made using programmed start and stop events that can be used with a set of predefined IG-XL measurement methods, allowing for easy time measurement calculations. The DCTime has two asynchronous trigger lines per DCTime allowing triggering of different instruments on the same DC instrument board using OnBoard mode and instruments on other boards using DIB Trigger Signal mode.

The following topics are available:

• [DCTime Overview](#)

• [Measurement Types](#)

• [Start Modes](#)

• [Enable Mode](#)

• [Interleave Mode](#)

• [Hysteresis](#)

• [Asynchronous Trigger](#)

• [DCTime Alarms](#)

For information on DCTime instrument performance specifications and standards, see:

• [Time Measurement \(DC30\)](#)

• [Time Measurement \(DC75\)](#)

DCTimeDefTopic

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About the DCTime

About the DCTime

The DC30 or DC75 DC Time Measurement Unit (DCTime) is a DC instrument that can perform time measurements, such as rise time, fall time, period, frequency and pulse width measurements. Time measurements are accurately made using programmed start and stop events, and a set of IG-XL measurement types allows for easy time measurement calculations.

Each DCTime instrument has two asynchronous trigger lines, allowing triggering of different instruments on the same DC instrument board using OnBoard mode and instruments on other boards using DIB Trigger Signal mode.

Pattern control can be used to program a DCTime, including microcode and PSet programming. The following topics are included:

- [Features](#)
- [Safety Information](#)
- [Key Terms](#)
- [DC30/75 DCTime Software Licensing Requirements](#)

The DCTime information includes the following sections:

- [DCTime Theory of Operation](#)
- [DCTime Programming](#)

- [DCTime Examples](#)

Related Topics

- [DC Debug Displays](#)
- [DC30 Specifications](#)
- [DC75 Specifications](#)
- [DCTime \(VBT syntax\)](#)
- [DC30 DIB Design Guidelines](#)
- [DC75 DIB Design Guidelines](#)

For more information on DC30 and DC75 instruments, see:

- [DC30 Instrument Board](#)
- [DC75 Instrument Board](#)
- [About the DCDiffMeter](#)
- [About the DCVI](#)

For information on UltraVI80 DCTime, see [About the UltraVI80 DCTime](#).

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About the DCTime : Features

Features

The DC30 or DC75 DC Time Measurement Unit includes the following features:

- | | |
|---|---|
| • | Voltage/impedance ranges |
| • | Two comparators |
| • | One attenuator |
| • | Input from channels sense lines |
| • | Dedicated input path per Time Front End |
| • | Trigger A/D conversion on any DCVI meter |
| • | Pattern control |
| • | Interleaving mode doubles the maximum input frequency from 20MHz to 40MHz |
| • | A dual time stamper |
| • | Both stampers of a DCTime share resources such as: |

- [Direct counter](#)

- [Clock](#)

- [Start signals](#)

- [Each time stamper has its own dedicated memory](#)

- [Each time stamper has four input signals:](#)

- [Event](#)

- [Enable](#)

- [Start](#)

- [Clock signals](#)

- [Both stampers in a DCTime share clock and start signals](#)

- [Each stamper has its own event signal](#)

For more information, see:

- [DC30 Specifications](#)

- [DC75 Specifications](#)



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About the DCTime : Safety Information

Safety Information

Follow general safety procedures when operating the DC instrument boards. Read all test system and instrument safety instructions before operating the tester. Only trained personnel should operate the test system. For more information, see [Test System Safety](#) in the [Safety](#) section.

General Warnings

Warning:

High voltages or currents or both may be present in the device interface. Contacting this area of the equipment can cause personal injury, damage to the device under test (DUT), or damage to external equipment.

Warning:

Even if a DC instrument is in a programmed state that is not forcing current or voltage, this does not imply any sense of safety. If a DC instrument is powered on, there is the potential of danger to the user at the DIB. For proper handling of a DIB, see [About Operator and Safety Information](#) in the [Safety](#) section.

Tester Cover

Keep the test head cover in place when the tester is in operation. If the test head cover is removed, an interlock is activated powering down the test system. Only trained and authorized personnel should operate the tester with the test head cover removed.

Warning:

Gate Off does not imply that no voltage exists at the channel output nor does it imply any sense of safety. Follow the proper safety rules before handling the DIB.

DIB Handling

DIB circuitry can store hazardous potential depending on the components on the DIB (such as capacitors). Take care when handling the DIB even after the test system is powered down. Since

DIBs are fabricated by customers, the test system cannot track if a DIB has safely dissipated stored energies.

 Warning:

Make sure that any power storage devices on a DIB have discharged stored potential before handling the DIB. Improper handling of a DIB can result in personnel injury.

		
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About the DCTime : Key Terms

Key Terms

DGS	Device ground sense is a sense signal that tracks the voltage of ground on the DIB.
DIB access	Allows a connection to be made to a DUT pin, connected to one instrument, by routing the single of the secondary instrument into the primary instrument’s DIB access line and to the pin through the primary instrument’s single path.
enable-event	Imposing a condition to be true before a time stamp can be taken.
event	An occurrence on an input signal when it passes through a specified voltage level (threshold) with a specified direction (slope).
interleave	A DCTime feature that allows alternate event detection and stamping by the first and second time stampers.
sense line	High impedance wiring used to measure the output of the force line. Also used to make connections to the meter for measurements on the DUT pin. This line is part of the control loop.
time stamp	The time an input signal event was detected.
XOR	An output from the DCTime Time Front End that goes to the DIB through a dedicated pin.

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[About the DCTime](#) : DC30/75 DCTime Software Licensing Requirements

[DC30/75 DCTime Software Licensing Requirements](#)
Operation of the DC30 and DC75 does not require a software license.
For more information, see [IG-XL Licensing](#) in IG-XL Administration help.

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DCTime Examples

DCTime Examples

This section contains the following kinds of program examples for the DC30 or DC75 DC Time Measurement Unit (DCTime):

- IG-XL workbook examples show all IG-XL data, VBT code, and patterns necessary for a complete running program.
- VBT function examples show specific applications of DC Instrument programming, such as measuring voltage and setting up a waveform.

The following topics are available:

- DCTime Rise Time Measurement Test
- Frequency Measurement with Sine Wave
- VBT Function Examples

Related Topics

- DCTime (VBT Syntax Reference)
- DCTime Programming

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DCTime Examples : DCTime Rise Time Measurement Test

DCTime Rise Time Measurement Test

This test makes a rise time measurement on a voltage regulator.

- Channel 1 of a DC30 board is connected to the input of a voltage regulator.
- Channel 2 of a DC30 board is connected to the output of a voltage regulator.
- Channel 1 is programmed to force a 3V ramp from 8V to 11V.
- Channel 2 is programmed to measure the output voltage.
- Channel 1 is programmed to measure the resulting rise time.

Required Programming Steps

- Set Up the DUT
- Set Up DataTool
- VBT Code

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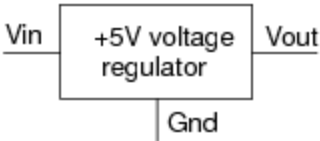
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DCTime Examples : DCTime Rise Time Measurement Test : Set Up the DUT

Set Up the DUT

This figure shows the DUT for this test.



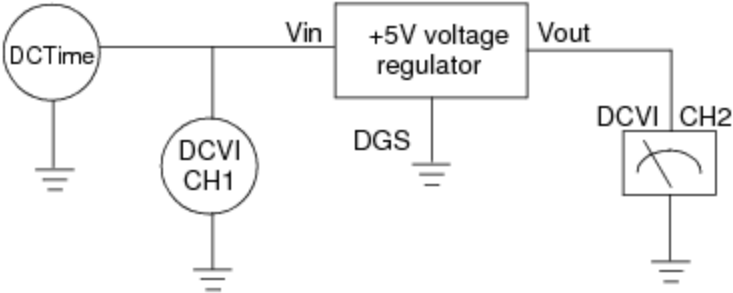
DUT Block Diagram

The table lists the specification for this test.

DUT Specification Sheet

Pin Name	Pin Function	Pin Description
Vin	Voltage input	Input voltage of the regulator.
Vout	Voltage output	Output voltage of the regulator.
Gnd	Ground	Regulator ground pin.

This figure shows DUT connections and DIB wiring required for this test.



DIB Layout

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DCTime Examples : [DCTime Rise Time Measurement Test](#) : Set Up DataTool

Set Up DataTool
The following DataTool sheets and data are used in this test:

- Pin Map Sheet
- Channel Map Sheet
- Test Instances Sheet
- Flow Table Sheet

Pin Map Sheet

Pin Map

Group Name	Pin Name	Type
	Vin	Analog
	Vout	Analog
	Ch1rise	Analog

Channel Map Sheet

Channel Map

DIB ID:

View Mode:

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Site 1
Vin		DCVI	16.sense1.dgs1	16.sense4.dgs1
Vout		DCVI	16.sense2.dgs2	16.sense4.dgs2
Ch1rise		DCTime	16.sense1	16.sense3

Test Instances Sheet

Test Instances

Test Procedure			
Test Name	Type	Name	Called As
SrcCapDCTime	VBT	DCTime_RiseTime	

Flow Table Sheet

Flow Table

Flow Domain:

Gate					Command	
Label	Enable	Job	Part	Env	Opcode	Parameter
					Test	SrcCapDCTime

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DCTime Examples : DCTime Rise Time Measurement Test : VBT Code

VBT Code

The following VBT code performs this test. You can copy and paste it into the VBT editor.

Public Function DCTime_RiseTime() As Long

' Connect the DCVIs

Call TheHdw.DCVI.Pins("Vin,Vout").Connect(tlDCVIConnectDefault)

' Create the samples for the ramp. Ramp is 8V-11V ramp

Dim VSamps As New DSPWave

Call VSamps.CreateRamp(8, 0.5, 7, DspDouble)

' Create the wave definition and source signal

Call TheExec.WaveDefinitions.CreateWaveDefinition("MyRampDef", VSamps, True)

With TheHdw.DCVI.Pins("Vin").Source.Signals.Add ("MyRamp")

.WaveDefinitionName = "MyRampDef"

End With

' Set up the source DCVI signal

With TheHdw.DCVI.Pins("Vin").Source.Signals.Item("MyRamp")

.Mode = tlDCVIModeVoltage

.Range.Value = 20

.Bandwidth = 50000

.SampleRate = 10000

.SampleSize = 7

.Amplitude = 1

.Offset = 0

.Phase = 0

.LoadSamples

```
.LoadSettings
End With
' Program the input pin
With TheHdw.DCVI.Pins("Vin")
    .Meter.Mode = tlDCVIMeterVoltage
    .Meter.Filter.Value = 5000
    .Meter.VoltageRange.Value = 10
    .Current = 0.1
    .Voltage = 8
    .VoltageRange.Autorange = True
    .Mode = tlDCVIModeVoltage
End With
' Program the channel connected to the output pin
With TheHdw.DCVI.Pins("Vout")
    .Meter.Mode = tlDCVIMeterVoltage
    .Meter.Filter.Value = 5000
    .Meter.VoltageRange.Value = 10
    .Current = -0.01
    .Voltage = 0
    .VoltageRange.Autorange = True
    .Mode = tlDCVIModeCurrent
End With
' Gate on DCVIs
TheHdw.DCVI.Pins("Vin,Vout").Gate = True
' Connect the DCTime
TheHdw.DCTime.Pins("ch1rise").Connect
' Set up the DCTime
With TheHdw.DCTime.Pins("ch1rise").Measurement.RiseTime
    .Hysteresis = tlDCTimeHysteresisOn
    .SetVoltageRange 20, tlDCTimeImpedanceHiZ
    .SetThresholds 8.3, 10.7
    .SampleSize = 200
End With
' Program a wait
TheHdw.Wait 0.025
' Create and program the capture DCVI signal
TheHdw.DCVI.Pins("Vout").Capture.Signals.Add ("MyCapture")
With TheHdw.DCVI.Pins("Vout").Capture.Signals.Item("MyCapture")
```

```

.Mode = tlDCVIMeterVoltage
.Range.Value = 5
.SampleRate = 10000
.SampleSize = 2000
.Filter.Value = 10000
.LoadSettings
End With
' Trigger signal
TheHdw.DCVI.Pins("Vout").Capture.Signals("MyCapture").Trigger
' DCTime Capture
Call TheHdw.DCTime.Pins("ch1rise").TimeStamps.Trigger ("DCTimeCapture")
' Start the source signal
Call TheHdw.DCVI.Pins("Vin").Source.Signals.Item("MyRamp").Start (tlStartImmediately)
' Program a wait
TheHdw.Wait 0.025
' Retrieve Stamps/Samples
Dim pld1 As New PinListData
Dim pld2 As New PinListData
Call TheHdw.DCTime.Pins("ch1rise").TimeStamps("DCTimeCapture").GetWaves(pld1, pld2)
' Program a wait
TheHdw.Wait 0.025
Dim Result_DSPWave As New DSPWave
Dim Result_RiseTime As New SiteDouble
Result_DSPWave = TheHdw.DCTime("ch1rise").Measurement.RiseTime _
    .CalculatedResults(pld1, pld2)
Dim Site As Variant
For Each Site In TheExec.Sites
    Result_RiseTime = Result_DSPWave.CalcMean
Next Site
' Compare results to limits in Flow Table
Call TheExec.Flow.TestLimit(resultVal:=Result_RiseTime, _
    ForceResults:=tlForceFlow)
' Disconnect and gate off the DCVIs
TheHdw.DCVI.Pins("Vin,Vout").Disconnect (tlDCVIConnectDefault)
TheHdw.DCVI.Pins("Vin,Vout").Gate = False
' Disconnect the DCTime
TheHdw.DCTime.Pins("ch1rise").Disconnect
End Function

```

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DCTime Examples : Frequency Measurement with Sine Wave

Frequency Measurement with Sine Wave

This example uses a DCVI to source a sine wave and a DCTime to capture the wave, then performs a frequency measurement on the captured time stamps and samples.

The DC30 board that sources the DCVI waveform and performs the DCTime capture is in slot 18.

The HSD1000 board that controls the pattern is in slot 16.

This workbook example includes the following program components:

- | | |
|---|-----------------|
| • | DataTool Sheets |
| • | VBT Code |
| • | DSP Code |

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DCTime Examples : Frequency Measurement with Sine Wave : DataTool Sheets

DataTool Sheets

This workbook example uses the following DataTool sheets:

- Pin Map Sheet
- Channel Map Sheet
- Time Sets (Basic) Sheet
- Pin Levels Sheet
- Test Instances Sheet
- Flow Table Sheet
- Global Specs

Pin Map Sheet

Pin Map

Group Name	Pin Name	Type
	DC1	Analog
	DCTime1	Analog
	hsd1	I/O

Channel Map Sheet

Channel Map

DIB ID:

View Mode: Signal

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Comm
DC1		DCVI	18.sense3.dgs2	
DCTime1		DCTime	18.sense3	
hsd1		I/O	16.ch0	

Time Sets (Basic) Sheet

Time Sets (Basic)

Timing Mode:

Single

Master Timeset Name:

Time Domain:

Strobe Ref Setup Name:

Time Set	Cycle	Pin/Group			Data		Drive				Compare				Edge Resolution
	Period	Name	Clock Period	Setup	Src	Fmt	On	Data	Return	Off	Mode	Open	Close	Ref Offset	Mode
tset_1us	1.E-06	hsd1			PAT	NR	0	0		1.E-06	Off				Machine

Pin Levels Sheet

Pin Levels

Pin/Group	Seq.	Parameter	Value
hsd1		Vil	0
hsd1		Vih	3
hsd1		Vol	0
hsd1		Voh	0
hsd1		VCmpMain	0
hsd1		Vt	0
hsd1		Vcl	=_Vcl_default
hsd1		Vch	=_Vch_default
hsd1		VoutLoTyp	0
hsd1		VoutHiTyp	0
hsd1		DriverMode	Largeswing-HiZ

Test Instances Sheet

Test Instances

Test Name	Test Procedure			DC Specs		AC Specs		Sheet Parameters				Mixed Signal Timing	Overlay
	Type	Name	Called As	Category	Selector	Category	Selector	Time Sets	Edge Sets	Pin Levels			
TI_Freq_canned_VBT	VBT	Frequency						TSB		Levels			

Flow Table Sheet

Flow Table

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					Flow Domain:													
					Gate		Command				Limits		Datalog Display		Bin Number		Sort Number	
Label	Enable	Job	Part	Env	Opcode	Parameter	TName	TNum	LoLim	HiLim	Scale	Units	Format	Pass	Fail	Pass	Fail	Result
					set-error-bin											99		99
					Test	TI_Freq_canned_VBT												
					Use-Limit	TI_Freq_canned_VBT	Freq_Average	1	900	1100					6		6	Fail
					Use-Limit	TI_Freq_canned_VBT	Freq_All	2	900	1100					6		6	Fail
					set-device									1		1		Pass
					stop													

Global Specs

Global Specs

Symbol	Job	Value	Comment
Vcl_default		- 1.E+00	Detector clamp voltage low
Vch_default		6.E+00	Detector clamp voltage high

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DCTime Examples : Frequency Measurement with Sine Wave : VBT Code

VBT Code

This example contains the following functions:

- **Function** Frequency sets up the DCVI and the DCTime, starts the sine wave, starts the capture, and analyzes and tests the captured data.
- **Function** CreateSignalsTest1 creates the sine wave that the DCVI sources.

Public Function Frequency() As Long

```
On Error GoTo errHandler
Dim pld1 As New PinListData
Dim pld2 As New PinListData
Dim AveragePLD As New PinListData
Dim AllPLD As New PinListData
Dim data As New DSPWave
Dim vSite As Variant
' Replace "DCTime1" with your DCTime pin name
Dim DCTimepinName As String
    DCTimepinName = "DCTime1"
' Replace "DC1" with your DCVI pin name
Dim DCVIpinName As String
    DCVIpinName = "DC1"
' Create a SinWave signal for DCVI pin
Call CreateSignalsTest1(DCVIpinName)
```

```

' Connect the DCVI channel and the DCTime channel to the DUT
TheHdw.DCVI.Pins(DCVIpinName).Connect (tlDCVIConnectDefault)
' Gate DCVI on
TheHdw.DCVI.Pins(DCVIpinName).Gate = True
' Connect the DCTime and set alarms to Default
TheHdw.DCTime.Pins(DCTimepinName).Connect
TheHdw.DCTime.Pins(DCTimepinName).Alarm(tlDCTimeAlarmAll) = tlAlarmDefault
' Set up the DCTime
TheHdw.DCTime.Pins(DCTimepinName).Mode = tlDCTimeModeStamper
TheHdw.DCTime.Pins(DCTimepinName).Interleave = tlDCTimeInterleaveOff
TheHdw.DCTime.Pins(DCTimepinName).Measurement.Frequency.Start.SetInput _
    tlDCTimeStartInputOnTrigger
With TheHdw.DCTime.Pins(DCTimepinName)
    .Stamper1.Holdoff.Count = 0
    .Stamper1.SampleSize = 10      ' The HoldOff is not programmed and
                                   ' the programmed Sample Size is 10
    .FrontEnd.SetAll 10, tlDCTimeImpedanceHiZ, 0.8, 0, tlDCTimeHysteresisOn
    .Stamper1.Event.Input = tlDCTimeInputThreshold1
    .Stamper1.Event.Slope = tlSlopePositive
End With

```

```

' Trigger the DCTime
TheHdw.DCTime.Pins(DCTimepinName).TimeStamps.Trigger ("my_capture")
' Start the sine wave
TheHdw.DCVI.Pins(DCVIpinName).Source.Signals("MySin").Start (tlStartImmediately)
' Wait time because 10 samples to be captured (Sine Period=1ms)
TheHdw.Wait 0.02
' Retrieve Stamps/Samples
TheHdw.DCTime.Pins(DCTimepinName).TimeStamps("my_capture").GetWaves pld1, pld2
TheHdw.Wait 0.0001

```

```

' Stop the sine wave
TheHdw.DCVI.Pins(DCVIpinName).Source.Signals.Stop (tlStopImmediately)
' Gate DCVI off
TheHdw.DCVI.Pins(DCVIpinName).Gate = False
TheHdw.Wait 0.0003 ' Wait for V=0 before disconnecting
' Disconnect the DCVI channel and the DCTime channel to the DUT
TheHdw.DCVI.Pins(DCVIpinName).Disconnect (tlDCVIConnectDefault)

```

```

' Disconnect the DCTime
TheHdw.DCTime.Pins(DCTimepinName).Disconnect
' -----
' Process Stamps/Samples to get an average
AveragePLD = TheHdw.DCTime.Pins( _
    DCTimepinName).Measurement.Frequency.CalculatedAverageResults( _
    pld1)
' or to get all samples
AllPLD = TheHdw.DCTime.Pins( _
    DCTimepinName).Measurement.Frequency.CalculatedResults( _
    pld1)
' If retrieving all samples, processing continues to get an average
data = AllPLD.Pins(DCTimepinName)
' Make the test passing if offline
If TheExec.TesterMode = testModeOffline Then
    AveragePLD = 1000
    For Each vSite In TheExec.Sites.Active
        data.FillWithConstant (1000)
    Next vSite
End If
' -----
' Compare measure with the limits
TheExec.Flow.TestLimit resultval:=AveragePLD, ForceResults:=tIfForceFlow
TheExec.Flow.TestLimit resultval:=data.CalcMean, ForceResults:=tIfForceFlow
Exit Function
errHandler:
    If AbortTest Then Exit Function Else Resume Next
End Function
Private Function CreateSignalsTest1(PinName As String)
    ' Create samples for a ramp
    Dim Voltages As New DSPWave
    Call Voltages.CreateSin(2 * Pi / 100, 0, 100)
    ' Create the ramp for the DCVI channel.
    ' The ramp sources the samples created above.
    TheExec.WaveDefinitions.CreateWaveDefinition "MySinDef", Voltages, True
    TheHdw.DCVI.Pins(PinName).Source.Signals.Add ("MySin")
    TheHdw.DCVI.Pins(PinName).Source.Signals("MySin").WaveDefinitionName = _
        "MySinDef"

```

```
' Program the source signal using the wave definition created earlier.  
With TheHdw.DCVI.Pins(PinName).Source.Signals("MySin")  
  .Mode = tIDCVIModeVoltage  
  .Range.Value = 5  
  .Bandwidth.Value = 50000  
  .Amplitude = 1  
  .Offset = 0  
  .Phase = 0  
  .SampleRate = 100000      ' 1kHz SinWave  
  .SampleSize = 100  
  .LoadSettings  
End With  
End Function
```

dctime_apps.4.09

	
---	---

[DCTime Examples : Frequency Measurement with Sine Wave : DSP Code](#)

DSP Code

This example does not use DSP procedures.

	
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dctime_apps.4.10

	
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DCTime Examples : VBT Function Examples

VBT Function Examples

The following VBT examples are included:

- | | |
|---|--|
| • | Rise Time Measurement |
| • | Fall Time Measurement |
| • | Frequency Measurement |
| • | Pulse Width |
| • | Period Measurement |
| • | Duty Cycle Measurements |
| • | Pin-to-Pin Propagation Delay Measurements |
| • | Digital-to-Analog Propagation Delay |
| • | Program Individual Parameters to Program a DCTime PSet |
| • | Program Set Statements to Program a DCTime PSet |

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dctime_apps.4.11



[DCTime Examples](#) : [VBT Function Examples](#) : Rise Time Measurement

Rise Time Measurement

[Public Function](#) RiseTime() As Long

[Dim](#) pld1 As New PinListData

Dim pld2 As New PinListData

' [Replace "DCTime1" with your DCTime pin name](#)

Dim pinName As String

pinName = "DCTime"

' [Connect the DCTime to the DUT](#)

TheHdw.DCTime.Pins(pinName).Connect

' [Set up the DCTime](#)

With TheHdw.DCTime.Pins(pinName).Measurement.RiseTime

.Start.SetInput tlDCTimeStartInputOnTrigger

.SetVoltageRange 10, tlDCTimeImpedanceHiZ

.SetThresholds 1, 4

.Hysteresis = tlDCTimeHysteresisOnPosAdjust

.SampleSize = 10

End With

' [Trigger the DCTime to make rise time measurement](#)

TheHdw.DCTime.Pins(pinName).TimeStamps.Trigger ("my_capture")

' [Retrieve Stamps/Samples](#)

TheHdw.DCTime.Pins(pinName).TimeStamps("my_capture").GetWaves pld1, pld2

[Dim](#) AveragePLD As New PinListData

Dim AllPLD As New PinListData

Dim AverageResults As Double

```
Dim AllResults As Double
Dim data As New DSPWave
' Process Stamps/Samples to get an average
AveragePLD = TheHdw.DCTime.Pins(pinName).Measurement.RiseTime _
    .CalculatedAverageResults(pld1, pld2)
' or to get all samples
AllPLD = TheHdw.DCTime.Pins(pinName).Measurement.RiseTime _
    .CalculatedResults(pld1, pld2)
' If retrieving an Average, processing continues
AverageResults = AveragePLD
' If retrieving all samples, processing continues to get an average
data = AllPLD
AllResults = data.CalcMean
End Function
```

dctime_apps.4.12

	
---	---

DCTime Examples : VBT Function Examples : Fall Time Measurement

Fall Time Measurement

Public Function FallTime() As Long

Dim pld1 As New PinListData

Dim pld2 As New PinListData

Dim pinName As String

pinName = "DCTime1" ' Replace "DCTime1" with your DCTime pin name

' Connect the DCTime and set alarms to Default

TheHdw.DCTime.Pins(pinName).Connect

TheHdw.DCTime.Pins(pinName).Alarm(tlDCTimeAlarmAll) = tlAlarmDefault

' Set up the DCTime

TheHdw.DCTime.Pins(pinName).Mode = tlDCTimeModeStamper

TheHdw.DCTime.Pins(pinName).Interleave = tlDCTimeInterleaveOn

TheHdw.DCTime.Pins(pinName).Measurement.FallTime.Start _

.SetInput tlDCTimeStartInputOnTrigger

With TheHdw.DCTime.Pins(pinName).Measurement.FallTime

.SetVoltageRange 10, tlDCTimeImpedanceHiZ

.SetThresholds 1.5, 0.5

.SampleSize = 100

.Hysteresis = tlDCTimeHysteresisOnNegAdjust

End With

' Trigger the DCTime

TheHdw.DCTime.Pins(pinName).TimeStamps.Trigger ("my_capture")

TheHdw.Wait 0.01

' Retrieve Stamps/Samples

TheHdw.DCTime.Pins(pinName).TimeStamps("my_capture").GetWaves pld1, pld2

```
Dim AveragePLD As New PinListData
Dim AllPLD As New PinListData
Dim AverageResults As Double
Dim AllResults As Double
Dim data As New DSPWave
' Process Stamps/Samples to get an average
AveragePLD = TheHdw.DCTime.Pins(pinName).Measurement.FallTime _
    .CalculatedAverageResults(pld1, pld2)
' or to get all samples
AllPLD = TheHdw.DCTime.Pins(pinName).Measurement.FallTime _
    .CalculatedResults(pld1, pld2)
' If retrieving an Average, processing continues
AverageResults = AveragePLD
' If retrieving all samples, processing continues to get an average
data = AllPLD.Pins
AllResults = data.CalcMean
End Function
```

dctime_apps.4.13

	
---	---

DCTime Examples : VBT Function Examples : Frequency Measurement

Frequency Measurement

Public Function Frequency() As Long

Dim pld1 As New PinListData

Dim pld2 As New PinListData

Dim pinName As String

pinName = "DCTime1" ' Replace "DCTime1" with your DCTime pin name

' **Connect the DCTime and set alarms to Default**

TheHdw.DCTime.Pins(pinName).Connect

TheHdw.DCTime.Pins(pinName).Alarm(tlDCTimeAlarmAll) = tlAlarmDefault

' **Set up the DCTime**

TheHdw.DCTime.Pins(pinName).Mode = tlDCTimeModeStamper

TheHdw.DCTime.Pins(pinName).Interleave = tlDCTimeInterleaveOff

TheHdw.DCTime.Pins(pinName).Measurement.Frequency.Start.SetInput _
tlDCTimeStartInputOnTrigger

With TheHdw.DCTime.Pins(pinName).Measurement.Frequency

.SetVoltageRange 10, tlDCTimeImpedanceHiZ

.Hysteresis = tlDCTimeHysteresisOn

.Threshold = 1

.SampleSize = 100

.Slope = tlSlopePositive

End With

' **Trigger the DCTime**

TheHdw.DCTime.Pins(pinName).TimeStamps.Trigger ("my_capture")

' **Retrieve Stamps/Samples**

TheHdw.DCTime.Pins(pinName).TimeStamps("my_capture").GetWaves pld1, pld2

Dim AveragePLD As New PinListData

Dim AllPLD As New PinListData

Dim AverageResults As Double

Dim AllResults As Double

Dim data As New DSPWave

' Process Stamps/Samples to get an average

Set AveragePLD = TheHdw.DCTime.Pins(pinName).Measurement.Frequency _
 .CalculatedAverageResults(pld1)

' or to get all samples

Set AllPLD = TheHdw.DCTime.Pins(pinName).Measurement.Frequency _
 .CalculatedResults(pld1)

' If retrieving an Average, processing continues

AverageResults = AveragePLD

' If retrieving all samples, processing continues to get an average

data = AllPLD.Pins

AllResults = data.CalcMean

End Function

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dctime_apps.4.14

	
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DCTime Examples : VBT Function Examples : Pulse Width

Pulse Width

Public Function PulseWidth() As Long

Dim pld1 As New PinListData

Dim pld2 As New PinListData

Dim pinName As String

pinName = "DCTime1" ' Replace "DCTime1" with your DCTime pin name

' **Connect the DCTime and set alarms to Default**

TheHdw.DCTime.Pins(pinName).Connect

TheHdw.DCTime.Pins(pinName).Alarm(tlDCTimeAlarmAll) = tlAlarmDefault

' **Set up the DCTime**

TheHdw.DCTime.Pins(pinName).Mode = tlDCTimeModeStamper

TheHdw.DCTime.Pins(pinName).Interleave = tlDCTimeInterleaveOn

TheHdw.DCTime.Pins(pinName).Measurement.PulseWidth.Start.SetInput

tlDCTimeStartInputOnTrigger

With TheHdw.DCTime.Pins(pinName).Measurement.PulseWidth

.SetVoltageRange 10, tlDCTimeImpedanceHiZ

.Threshold = 1

.SampleSize = 100

.Slope = tlSlopePositive

.Hysteresis = tlDCTimeHysteresisOn

End With

' **Trigger the DCTime**

TheHdw.DCTime.Pins(pinName).TimeStamps.Trigger ("my_capture")

' **Retrieve Stamps/Samples**

TheHdw.DCTime.Pins(pinName).TimeStamps("my_capture").GetWaves pld1, pld2

Dim AveragePLD As New PinListData

Dim AllPLD As New PinListData

Dim AverageResults As Double

Dim AllResults As Double

Dim data As New DSPWave

' Process Stamps/Samples to get an average

AveragePLD = TheHdw.DCTime.Pins(pinName).Measurement.PulseWidth _
 .CalculatedAverageResults(pld1, pld2)

' or to get all samples

AllPLD = TheHdw.DCTime.Pins(pinName).Measurement.PulseWidth _
 .CalculatedResults(pld1, pld2)

' If retrieving an Average, processing continues

AverageResults = AveragePLD

' If retrieving all samples, processing continues to get an average

data = AllPLD.Pins(pinName)

AllResults = data.CalcMean

End Function

dctime_apps.4.15

	
---	---

DCTime Examples : VBT Function Examples : Period Measurement

Period Measurement

Public Function Period() As Long

Dim pld1 As New PinListData

Dim pld2 As New PinListData

Dim pinName As String

pinName = "DCTime1" ' Replace "DCTime1" with your DCTime pin name

' Connect the DCTime and set alarms to Default

TheHdw.DCTime.Pins(pinName).Connect

TheHdw.DCTime.Pins(pinName).Alarm(tlDCTimeAlarmAll) = tlAlarmDefault

' Set up the DCTime

TheHdw.DCTime.Pins(pinName).Mode = tlDCTimeModeStamper

TheHdw.DCTime.Pins(pinName).Interleave = tlDCTimeInterleaveOn

TheHdw.DCTime.Pins(pinName).Measurement.Period.Start.SetInput tlDCTimeStartInputOnTrigger

With TheHdw.DCTime.Pins(pinName).Measurement.Period

.SetVoltageRange 10, tlDCTimeImpedanceHiZ

.Threshold = 1

.SampleSize = 100

.Slope = tlSlopePositive

.Hysteresis = tlDCTimeHysteresisOn

End With

' Trigger the DCTime

TheHdw.DCTime.Pins(pinName).TimeStamps.Trigger ("my_capture")

' Retrieve Stamps/Samples

TheHdw.DCTime.Pins(pinName).TimeStamps("my_capture").GetWaves pld1, pld2

```
Dim AveragePLD As New PinListData
Dim AllPLD As New PinListData
Dim AverageResults As Double
Dim AllResults As Double
Dim data As New DSPWave
' Process Stamps/Samples to get an average
Set AveragePLD = TheHdw.DCTime.Pins(pinName).Measurement.Period _
    .CalculatedAverageResults(pld1, pld2)
' or to get all samples
Set AllPLD = TheHdw.DCTime.Pins(pinName).Measurement.Period _
    .CalculatedResults(pld1, pld2)
' If retrieving an Average, processing continues
AverageResults = AveragePLD
' If retrieving all samples, processing continues to get an average
data = AllPLD
AllResults = data.CalcMean
End Function
```

dctime_apps.4.16

	
---	---

DCTime Examples : VBT Function Examples : Duty Cycle Measurements

Duty Cycle Measurements

Public Function DutyCycle() As Long

Dim pld1 As New PinListData

Dim pld2 As New PinListData

Dim pinName As String

pinName = "DCTime1" ' Replace "DCTime1" with your DCTime pin name

' **Connect the DCTime and set alarms to Default**

TheHdw.DCTime.Pins(pinName).Connect

TheHdw.DCTime.Pins(pinName).Alarm(tlDCTimeAlarmAll) = tlAlarmDefault

' **Set up the DCTime**

TheHdw.DCTime.Pins(pinName).Mode = tlDCTimeModeStamper

TheHdw.DCTime.Pins(pinName).Interleave = tlDCTimeInterleaveOn

TheHdw.DCTime.Pins(pinName).Measurement.DutyCycle.Start.SetInput

tlDCTimeStartInputOnTrigger

With TheHdw.DCTime.Pins(pinName).Measurement.DutyCycle

.SetVoltageRange 10, tlDCTimeImpedanceHiZ

.Threshold = 1

.SampleSize = 100

.Slope = tlSlopePositive

.Hysteresis = tlDCTimeHysteresisOn

End With

' **Trigger the DCTime**

TheHdw.DCTime.Pins(pinName).TimeStamps.Trigger ("my_capture")

TheHdw.Wait 0.01

' Retrieve Stamps/Samples

TheHdw.DCTime.Pins(pinName).TimeStamps("my_capture").GetWaves pld1, pld2

Dim AveragePLD As New PinListData

Dim AllPLD As New PinListData

Dim AverageResults As Double

Dim AllResults As Double

Dim data As New DSPWave

' Process Stamps/Samples to get an average

Set AveragePLD = TheHdw.DCTime.Pins(pinName).Measurement.DutyCycle _
.CalculatedAverageResults(pld1, pld2)

' or to get all samples

Set AllPLD = TheHdw.DCTime.Pins(pinName).Measurement.DutyCycle _
.CalculatedResults(pld1, pld2)

' If retrieving an Average, processing continues

AverageResults = AveragePLD

' If retrieving all samples, processing continues to get an average

data = AllPLD

AllResults = data.CalcMean

End Function

dctime_apps.4.17

	
---	---

DCTime Examples : VBT Function Examples : Pin-to-Pin Propagation Delay Measurements

Pin-to-Pin Propagation Delay Measurements

Public Function DCTime_PropDelay() As Long

Dim pld1 As New PinListData

Dim pld2 As New PinListData

' When performing a pin-to-pin delay measurement, use both DCTime units since

' a single DCTime unit has only a single front end

TheHdw.DCTime.Pins("DCTime_Pins").Connect

TheHdw.DCTime.Pins("DCTime_Pins").Alarm(tlDCTimeAlarmAll) = tlAlarmDefault

' Program the DCTime to connect the Stamper to the output of the comparator

TheHdw.DCTime.Pins("DCTime_Pins").Mode = tlDCTimeModeStamper

' When programming DCTime to perform a pin-to-pin measurement, the DCTime Unit

' only uses Stamper1. As a result, you want to ensure that Interleave mode is

' Off so that Stamper1 is not waiting for Stamper2 to perform a measurement.

TheHdw.DCTime.Pins("DCTime_Pins").Interleave = tlDCTimeInterleaveOff

' To synchronize the time stamps taken by the 2 DCTime Units, you must use the

'.Start or .Enable signal to ensure that the Stampers take the time stamps in

' the order you expect. Use the .Start parameter and use whatever is

' considered to be the "second" signal as your .Start Input.

TheHdw.DCTime.Pins("DCTime_Pins").Measurement.PinToPinDelay.Start.SetInput

tlDCTimeStartInputPin, "DCTime2", "DCTime", 1

TheHdw.DCTime.Pins("DCTime_Pins").Measurement.PinToPinDelay.Start.Slope = tlSlopeNegative

' Programming Statements for the DCTime

With TheHdw.DCTime.Pins("DCTime_Pins").Measurement.PinToPinDelay

.SetVoltageRange 10, tlDCTimeImpedanceHiZ

.Threshold = 1

```
.SampleSize = 100
.Slope = tlSlopePositive
.Hysteresis = tlDCTimeHysteresisOn
End With
' This triggers BOTH DCTime Units. After this command is executed and all
' .Start Input conditions are met, any time the input signal meets the
' Threshold condition, and the Sample Size has not been met, the DCTime Unit
' will "stamp" for a measurement.
TheHdw.DCTime.Pins("DCTime_Pins").TimeStamps.Trigger ("my_capture")
TheHdw.Wait 0.01
Dim AveragePLD As New PinData
Dim AllPLD As New PinData
Dim AverageResults As Double
Dim AllResults As Double
Dim data As New DSPWave
' Retrieve the samples out of memory
TheHdw.DCTime.Pins("DCTime_Pins").TimeStamps("my_capture").GetWaves pld1, pld2
TheHdw.Wait 0.001
' Process the samples to provide a pin-to-pin measurement.
AllPLD = TheHdw.DCTime.Pins("DCTime_Pins").Measurement.PinToPinDelay. _
CalculatedResults(pld1.Pins("DCTime1"), pld1.Pins("DCTime2"))
' If retrieving all samples, processing continues to get an average
data = AllPLD
AllResults = data.CalcMean
End Function
```

dctime_apps.4.18

	
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DCTime Examples : VBT Function Examples : Digital-to-Analog Propagation Delay

Digital-to-Analog Propagation Delay

This example measures the time between a pattern microcode (digital) and an analog event coming from the DUT into DCTime1.

Public Function PropagationDelay()

' Replace "DCTime1" below with your DCTime pin name

Dim pinName As String

pinName = "DCTime1"

' Connect the DCTime

TheHdw.DCTime.Pins(pinName).Connect

With TheHdw.DCTime.Pins(pinName)

.Mode = tIDCTimeModeStamper

.FrontEnd.VoltageRange.SetValue 10, tIDCTimeImpedanceHiZ

.FrontEnd.Hysteresis = tIDCTimeHysteresisOn

.FrontEnd.Threshold1 = 2.5

.Interleave = tIDCTimeInterleaveOn

.Start.SetInput tIDCTimeStartInputOnTrigger

' A time stamp occurs on a "Stamp" microcode from the Pattern.

.Stamper1.Event.Input = tIDCTimeInputPattern

' Stamper 1 will be enabled from the DCTime Trigger Command

.Stamper1.Enable.SetInput tIDCTimeEnableInputNone

.Stamper1.SampleSize = 1

.Stamper1.Holdoff.Mode = tIDCTimeHoldOffModeOff

.Stamper2.Event.Input = tIDCTimeInputThreshold1

.Stamper2.Event.Slope = tISlopePositive

' Stamper 2 will be enabled from the DCTime Trigger Command

```
.Stamper2.Enable.SetInput tIDCTimeEnableInputNone
.Stamper2.SampleSize = 1
.Stamper2.Holdoff.Mode = tIDCTimeHoldOffModeOff
```

End With

' This Triggers BOTH DCTime Units. After this command is executed and all
' .Start Input conditions are met, any time the input signal meets the
' Threshold condition, and the Sample Size has not been met, the DCTime Unit
' will "stamp" for a measurement.

```
TheHdw.DCTime.Pins(pinName).TimeStamps.Trigger ("my_capture")
```

End Function

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[DCTime Examples](#) : [VBT Function Examples](#) : Program Individual Parameters to Program a DCTime PSet

[Program Individual Parameters to Program a DCTime PSet](#)

[Public Function DCTimePSet\(\)](#)

[' Replace "DCTime1" with your DCTime pin name](#)

Dim pinName As String

pinName = "DCTime1"

[' Create a PSet](#)

TheHdw.DCTime.Pins(pinName).PSets.Add ("PSetName")

[' Program individual parameters](#)

With TheHdw.DCTime.Pins(pinName).PSets("PSetName")

.FrontEnd.VoltageRange.SetValue 10, tIDCTimeImpedanceHiZ

.FrontEnd.Threshold1 = 5#

.FrontEnd.Threshold2 = 1#

.FrontEnd.Hysteresis = tIDCTimeHysteresisOn

.Stamper1.Event.Input = tIDCTimeInputThreshold1

.Stamper1.Event.Slope = tISlopePositive

.Stamper1.Enable.SetInput tIDCTimePSetEnableInputNone

.Stamper1.Holdoff.Mode = tIDCTimeHoldOffModeOff

.Stamper2.Event.Input = tIDCTimeInputThreshold1

.Stamper2.Event.Slope = tISlopePositive

.Stamper2.Enable.SetInput tIDCTimePSetEnableInputNone

.Stamper2.Holdoff.Mode = tIDCTimeHoldOffModeOff

.Start.SetInput (tIDCTimePSetStartInputOnTrigger)

.Interleave = tIDCTimeInterleaveOn

End With

' Apply the PSet

TheHdw.DCTime.Pins("Pin").PSets("PSetName").Apply

End Function

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[DCTime Examples](#) : [VBT Function Examples](#) : Program Set Statements to Program a DCTime PSet

[Program Set Statements to Program a DCTime PSet](#)

[Public Function DCTimeSetStatement\(\)](#)

' [Replace "DCTime1" with your DCTime pin name](#)

Dim pinName As String

pinName = "DCTime1"

' [Create a PSet](#)

TheHdw.DCTime.Pins(pinName).PSets.Add ("PSetName")

' [Program with Set statement](#)

```
TheHdw.DCTime.Pins(pinName).PSets("PSetName").Set 10, tlDCTimeImpedanceHiZ _
, tlDCTimeHysteresisOff, tlDCTimeHysteresisOn, 5#, 1# _
, tlDCTimeInterleaveOn, tlDCTimePSetStartInputOnTrigger, "", "", 1 _
, tlSlopePositive, tlDCTimePSetEnableInputNone, "", "", 1 _
, tlDCTimeEnableModeTriggerEach, tlSlopePositive
, tlDCTimeInputThreshold1, tlSlopePositive, 1, tlDCTimeHoldOffModeOff _
, 1, 10, tlDCTimePSetEnableInputNone, "", "", 1 _
, tlDCTimeEnableModeTriggerEach, tlSlopePositive
, tlDCTimeInputThreshold2, tlSlopePositive, 1, tlDCTimeHoldOffModeOff _
, 1, 10
```

' [Apply the PSet](#)

TheHdw.DCTime.Pins("Pin").PSets("PSetName").Apply

[End Function](#)

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DCTime Programming

DCTime Programming

This section describes programming of a DC Time Measurement Unit for DC30 and DC75 instrument boards.

The following topics are available:

- DCTime DataTool Programming
- Connecting a DCTime
- Programming Asynchronous Trigger Measurements
- DCTime Stamp Microcode Latency
- Visual Basic for Test
- DCTime Reset Values
- DCTime Pattern Control

For more information, see:

- DCTime (VBT Syntax Reference)

•	DCTime Examples
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DCTime Programming : DCTime DataTool Programming

DCTime DataTool Programming

This section provides instructions on setting up parameters in worksheets specific to the DCTime. The following topics are included:

- [Programming the Pin Map](#)
- [Programming the Channel Map](#)
- [DIB Access Channel Map Setup: DCTime as a Primary Instrument](#)

For more information, see [About DataTool](#).

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DCTime Programming : [DCTime DataTool Programming](#) : Programming the Pin Map

Programming the Pin Map

A pin map describes the connections from the tester to the device under test. A Pin Map sheet is required in a test program. Pin type determines what is programmable in the levels sheet. The recommended Type selection for the DCTime instrument is:

Analog	This type is used for the DCTime instrument.
--------	--

For more information, see [Pin Map Sheet](#) in the DataTool section.

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DCTime Programming : [DCTime DataTool Programming](#) : Programming the Channel Map

Programming the Channel Map

A DC30/75 instrument board can support 250 sites. For each pin that is connected to a DCTime, an entry is required in the PinName, Type, and each Site column. Type selections for DCTime instrument channels are:

DCTime	Sense line DIB address connection to a DCTime.
DCTimeHP	High performance input channel. This line avoids the DCVI path, improving signal quality.
DCTimeTrig	The trigger output that allows the DCTime to enable or start another instrument.

For more information, see [Channel Map Sheet](#) in the DataTool section.

Site Column Entries for DC30

In the Site columns, the channel map is programmed with the DIB connector number, the sense line and optionally the DGS connection. If multiplexed channels are available the mux side indicator is also required. The DIB connector location is dependent on tester configuration. The following table shows an example of valid entries:

Example of a DCTime Channel Map			
Pin Name	Type	Site 0	Comment
Pin01	DCTime	23.sense1	DC30 channel 1 in DIB slot 23 using default dgs.

Pin02	DCTime	50.sense1a.dgs2	DC75 channel 1 in DIB slot 50, A side of Mux using dgs2.
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This information can be seen in the file <installdir>\tester\CurrentChannelMap.txt.

Site Column Entries for DC75

The DC75 supports internal multiplexed channels allowing more connections to resources without taking up DIB space. Select multiplexed channels by entering the channel DIB address for the mux you wish to connect to. The table lists the channel DIB address for multiplexed channels.

Channel DIB Address for Multiplexed Channels

DC75 Channel	A Mux	B Mux
Channel 1	23.sense1a	23.sense1b
Channel 2	23.sense2a	23.sense2b
Channel 3	23.sense3a	23.sense3b
Channel 4	23.sense4a	23.sense4b

In the Site columns, the channel map is programmed with the DIB connector number, the sense line with the mux side indicator and optionally the DGS connection. The DIB connector location is dependant on tester configuration. The following table shows an example of valid entries:

DC75 Channel Map Entries

Pin Name	Type	Site 0	Comment
MyPin01	DCTime	50.sense1b	Channel 1 at DIB slot 50 using the default dgs.
MyPin02	DCTime	50.sense2a.dgs2	Channel 2 at DIB slot 50 using dgs 2.

This information is in the file \\<installdir>\tester\CurrentChannelMap.txt.

To have multiplexed connections, proper DIB layout is required. See DC75 Multiplexed Channel Mapping in the DIB Design Guidelines.

DIB Access Channel Map Setup: DCTime as a Primary Instrument

DIB access for the DCTime is similar to other UltraFLEX instruments. For more information, see:

- DC30 DIB Design Guidelines

- DC75 DIB Design Guidelines

An instrument with a connection to a DUT pin can be used as a primary instrument in a DIB access connection. The table shows an example of a channel map set up for DIB access using the DCTime as the primary instrument and a Digital pin as the secondary instrument.

Example Channel Map of a DCTime as the Primary Instrument

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Comment
Pin01		DCTime	32.hisense1a	Channel 1 low sense A side of Mux
DigPin:DA		IO	1.ch1	Digital using DCTime for DIB access

DIB Access Channel Map Setup: DCTime as a Secondary Instrument

An instrument that needs to connect to a DUT pin that another instrument is connected to can be used as a secondary instrument in a DIB access connection. The table shows an example of a channel map set up for DIB access using the DCTime as the secondary instrument and the Digital as the primary instrument.

Example Channel Map of a DCTime as the Secondary Instrument

Device Under Test		Tester Channel		
Pin Name	Package Pin	Type	Site 0	Comment
DigPin		IO	1.ch1	Digital pin connected to a DUT pin
DigPin:DA		DCTime	23.sense1	Channel 1 sense connected to digital pin DIB access pin

Note:

DIB access is not allowed between two resources assigned to the same pin name that use the same channel type. A DC30 DCTime cannot use a DC75 DCTime through DIB access since the channels use the same channel type.

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DCTime Programming : Connecting a DCTime

Connecting a DCTime

You can connect a DCTime to the DUT pin using VBT language. Connections can be made into the DCTime through one of the channel sense lines or using the High Performance Input. The DCTime can also connect to DIB Access. The channel connected depends on the channel map Type specified in the channel map.

The following connections can be made:

- DCTime Connected Through the Channel Sense
- DCTime Connected Through the High Performance Input
- DCTime Trigger Connected to the DUT

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DCTime Programming : Connecting a DCTime : DCTime Connected Through the Channel Sense

DCTime Connected Through the Channel Sense

The following channel map and VBT syntax connects the DCTime to the DUT through channel 1 sense line:

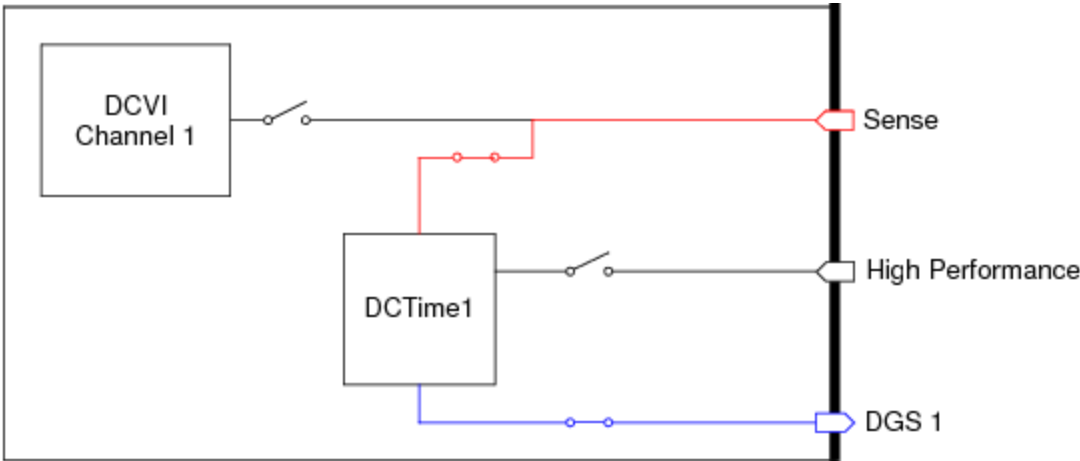
Example DCTime Channel Map

Pin Name	Type	Site 0	Comment
MyDC30Pin	DCTime	23.sense1.dgs1	DC30 channel
MyDC75Pin	DCTime	17.sense1a	DC75 channel 1 sense A side of Mux

' Connect the channel sense line

TheHdw.DCTime.Pins("MyDC30Pin").Connect

The figure shows the connection made:



DCTime Connection to DUT Through Channel Sense

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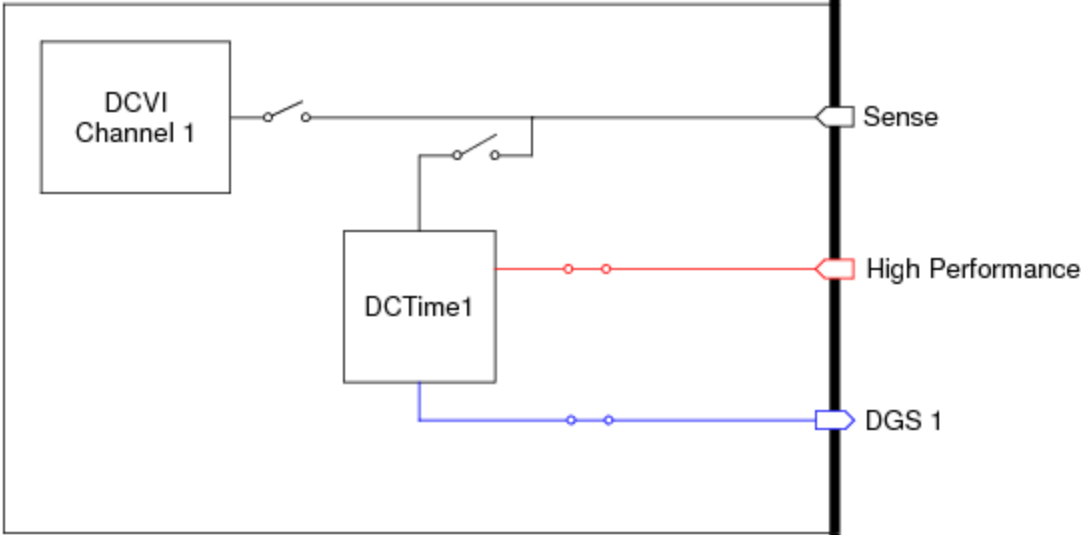
DCTime Programming : [Connecting a DCTime](#) : DCTime Connected Through the High Performance Input

[DCTime Connected Through the High Performance Input](#)
The following channel map and VBT syntax connects the DCTime to the DUT through the high performance Input:

Example DCTime Channel Map

Pin Name	Type	Site 0	Comment
MyDC30Pin	DCTimeHP	23.dig1.dgs1	DC30 high performance input 1
MyDC75Pin	DCTimeHP	17.dig1.dgs1	DC75 high performance input 1

' [Connect the channel sense line](#)
TheHdw.DCTime.Pins("MyDC75Pin").Connect
The figure shows the connection made:



DCTime Connection to DUT Through the High Performance Input

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DCTime Programming : Connecting a DCTime : DCTime Trigger Connected to the DUT

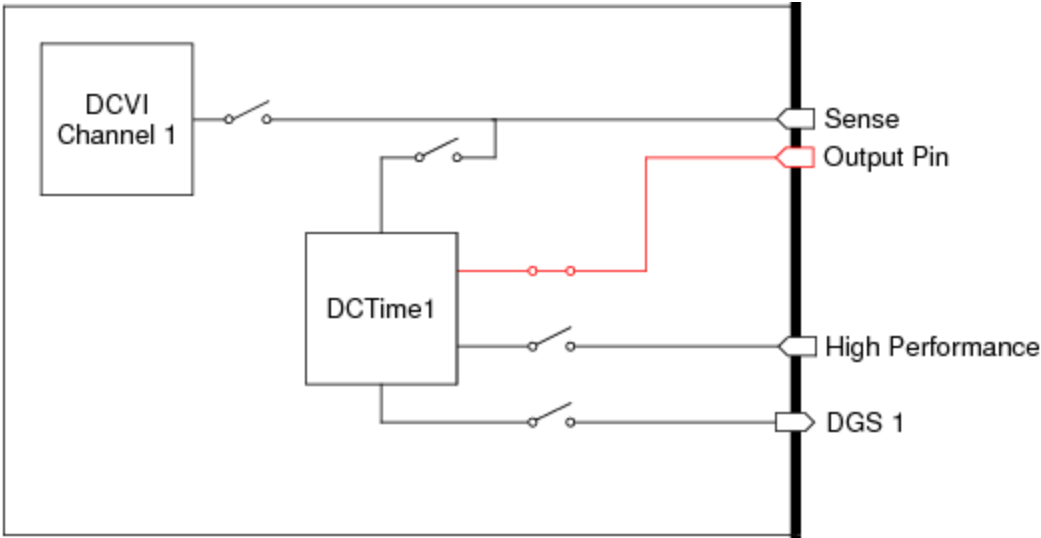
DCTime Trigger Connected to the DUT

The following channel map and VBT syntax connects the DCTime to the DIB through the DCTime Output pin:

Example DCTime Channel Map

Pin Name	Type	Site 0	Comment
MyDC30Pin	DCTimeTrig	23.dibaccess1	DC30 trigger line
MyDC75Pin	DCTimeTrig	17.dibaccess1	DC75 trigger line

TheHdw.DCTime.Pins("MyDC75Pin").Connect
The figure shows the connection made:



DCTime Trigger Connection to DIB Through DIB Access

☒ **Note:**

On the DC30 and DC75, the DCTime output connects to the DIB through DIB Access pins.

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DCTime Programming : Programming Asynchronous Trigger Measurements

Programming Asynchronous Trigger Measurements

A DCTime can asynchronously trigger instruments on the same DC instrument board using OnBoard trigger mode or can trigger instruments on other instrument boards using DIB Trigger mode.

Asynchronous triggering allows you to measure the state change of a device pin. Making an asynchronous trigger measurement requires both the DCTime front end comparator and a stamper. The front end comparator outputs a signal to the stamper that places a signal onto a trigger line connected to the specified meter. The types of asynchronous triggering are:

- | | |
|---|--------------------|
| • | OnBoard Triggering |
|---|--------------------|
- | | |
|---|----------------|
| • | DIB Triggering |
|---|----------------|

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DCTime Programming : Programming Asynchronous Trigger Measurements : OnBoard Triggering

OnBoard Triggering

An onboard asynchronous trigger allows you to trigger a DCVI meter or DCDiffMeter on the occurrence of an event using the DCTime instrument.

For more information, see:

- Onboard Asynchronous Triggering of a DCVI Meter
- Onboard Asynchronous Triggering of a DCDiffMeter

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DCTime Programming : Programming Asynchronous Trigger Measurements : DIB Triggering

DIB Triggering

DIB Triggering allows for a DCTime to asynchronously trigger an instrument on a different instrument board or a DUT through a DIB connection. The DCTime has one DIB Trigger line, per DCTime. For more information, see [DIB Triggering](#).

Channel Map

The channel map for DIB triggering requires that the channel map Type is set to DCTimeTrig as shown in [DCTime Trigger Connected to the DUT](#).

Programming a Trigger Pin

The DCTime listens for an event and on receiving that event sends a signal through the DIB trigger line to the instrument programmed to listen to it. The following are required to set up DCTime:

DIBTrigger	This mode notifies the DCTime to send the trigger signal through the DIB Triggering output. The DCTime listens on its inputs for an event to cross its threshold. Once this occurs it sends a trigger signal to the instrument programmed to listen to the trigger line. Set this using the following syntax: TheHdw.DCTime.Pins("MyPin").Mode = tIDCTimeModeDIBTrigger
Enabling the Trigger Pin	The asynchronous trigger must also be enabled. Enable the trigger pin using the following syntax. TheHdw.DCVI.Pins("MyPin").AsynchronousTrigger.Enable = True

You need not set the trigger pin slope for DCTime.
You must program the instrument that is waiting for the asynchronous trigger from the DCTime to receive the signal.

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DCTime Programming : DCTime Stamp Microcode Latency

DCTime Stamp Microcode Latency

The DCTime stamp microcode has a 320ns latency with respect to T0. You must account for this if the latency is a significant percentage of the measurement results. An example of this is making a pin-to-pin propagation delay measurement with a T0 less than 20s.

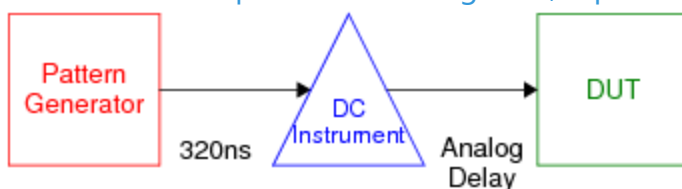
The following details this latency:

- [Hardware Microcode Latency](#)
- [Understanding Latency in a Pattern Vector](#)
- [Latency in a Digital to DC Propagation Measurement](#)

For more information, see [DC30 Specifications](#) or [DC75 Specifications](#).

Hardware Microcode Latency

All DC microcodes are designed to occur at the start of the T0 period, but because of the hardware latency this does not occur. There is a 320ns hardware delay between a microcode being issued at the start of T0 to when it actually occurs. The figure shows the latency in the hardware. Beyond the hardware delay of 320ns there is the possibility of a signal having additional delay because of instrument set up such as settling time, represented as analog delay.



Hardware Latency

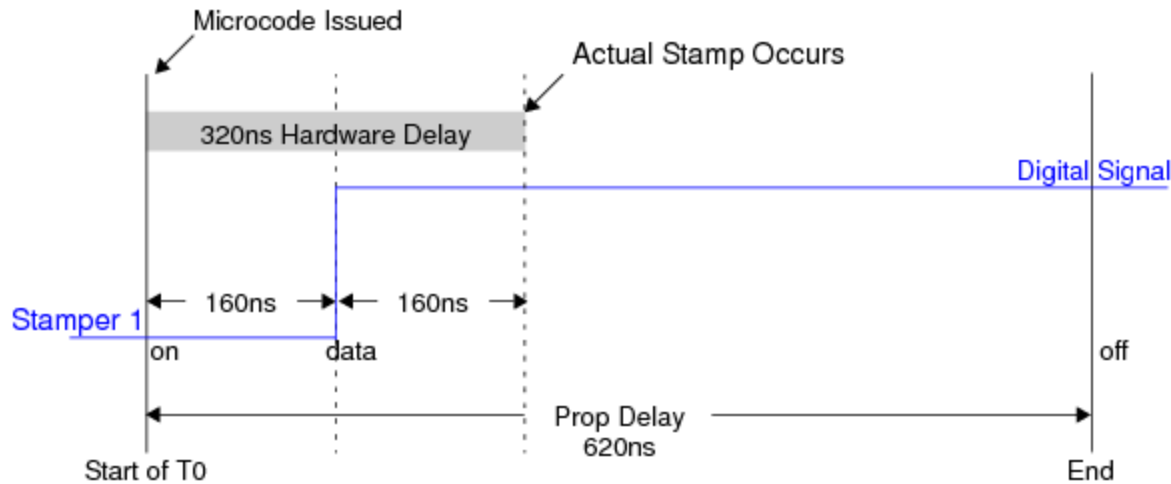
☒ **Note:**

All microcodes have this delay time, although test setup determines if the results are affected by the delay. See the hardware specification for details.

Understanding Latency in a Pattern Vector

In the following figure the tester period is 620ns. The timing for the digital signal is programmed such that the 0 to 1 transition occurs at data, which is set to 160ns. The time stamp occurs at the end of the latency delay rather than at the beginning of the T0 cycle.

The data transition on the digital pin was programmed to happen at 160ns, and the stamp microcode has a 320ns latency or hardware delay. Simple math determines that the stamp actually occurred $320 - 160 = 160$ ns after the digital edge.

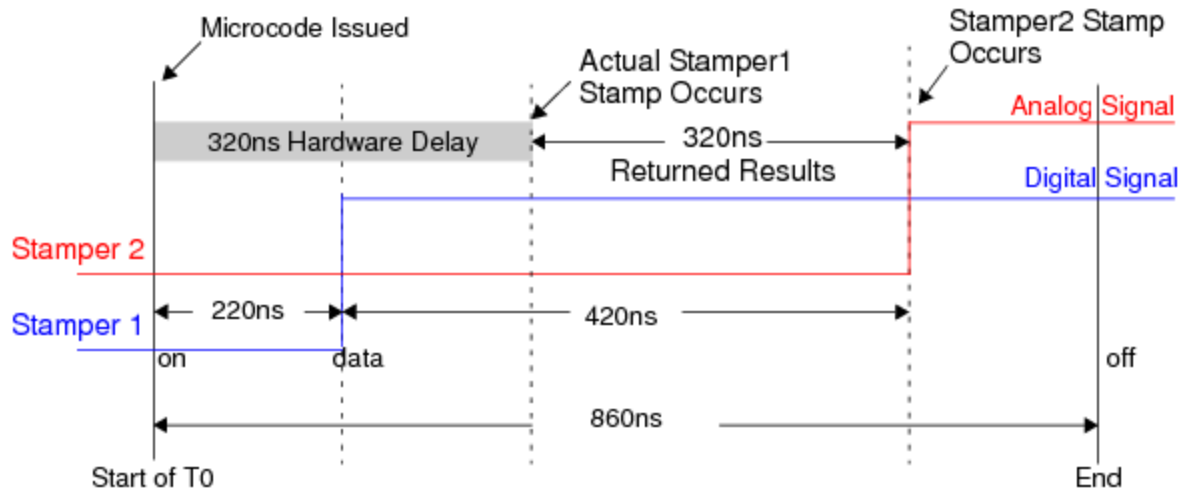


Latency in a Pattern

Latency in a Digital to DC Propagation Measurement

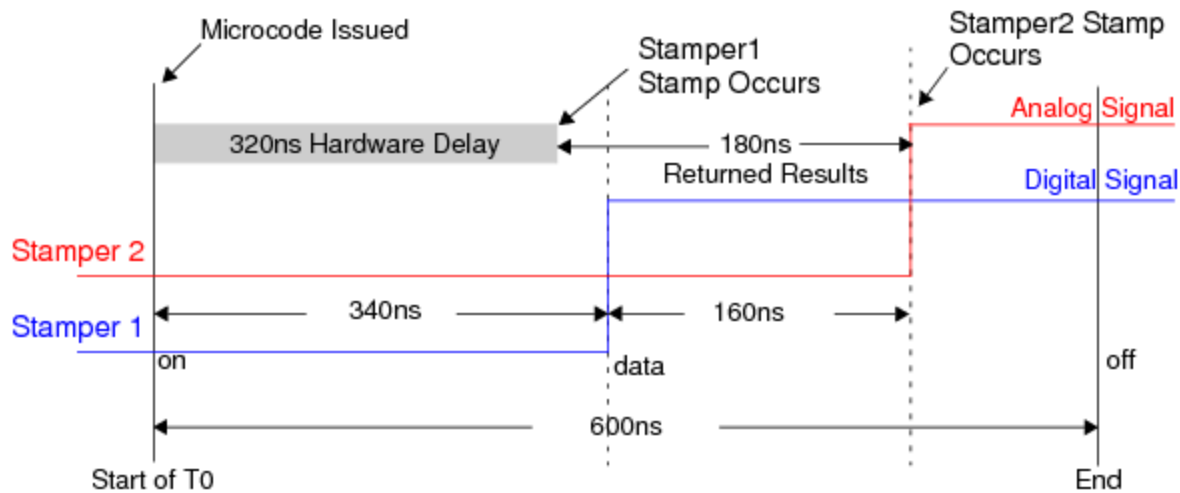
The following figure shows a pin-to-pin propagation test between a digital pin and an analog pin. DCTestime stamper1 is used to stamp the edge on the digital pin and stamper2 is triggered by the edge of the analog signal. DCTestime stamper1 stamps on the same vector as the digital pin changes state.

Since there is a delay of 320ns from the start of T0 and the time the Stamp is performed, the recorded time is off. In this example, stamper1 stamps at 320ns, because of the hardware delay. Stamper2 stamps at 640ns after T0 start, resulting in a propagation delay measurement of 320ns due to the hardware delay.



Digital-to-Analog Propagation Delay Setup, Data Less Than 320ns

In the following figure data is programmed to 340ns which is longer than the hardware delay. Since stamper1 stamps at the end of the hardware delay rather than at the start of T0 the returned results are longer than the expected value.



Digital-to-Analog Propagation Delay Setup, Data Greater Than 320ns

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DCTime Programming : Visual Basic for Test

Visual Basic for Test

DC instrumentation programs can be created using VBT language in the VBT editor. Each instrument has its own properties and methods used for controlling the functionality of the instrument.

For more information, see [DCTime](#) in the VBT Syntax Reference.

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DCTime Programming : DCTime Reset Values

DCTime Reset Values

Resetting the connections disconnects the DCTime from the DUT. Resetting the settings clears the DCTime memory and returns the DCTime to the default settings.

Perform a reset using the following VBT syntax:

TheHdw.DCTime.Pins(PinList).Reset(tlResetConnections Or tlResetSettings)

The table lists the reset state of DCTime parameters after a reset occurs.

DCTime Parameter Reset Values

DCTime Parameter	Reset Value
Holdoff	
Mode	Off
Count	0
Time	0
Hysteresis	Off
Measurement	Event Capture
Enable	
Input	None
Mode	Gate window
Slope	Positive
Start	

	Input	On trigger
	Slope	Positive
Stamper		
	Input	None
	Sample size	1
	Slope	Positive
Threshold		0V
Voltage range, impedance		0V, None

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DCTime Programming : DCTime Pattern Control

DCTime Pattern Control

A pattern is used to precisely program an instrument at the vector level. You can program DCTime parameters using microcode or by applying PSets. The following tools are used in pattern creation. You create a pattern independently of the test program using the PatternTool or a text editor. Patterns are composed of a series of numbered digital vectors. These digital vectors define the data for each channel listed in the pattern, as well the pattern microcode used for controlling pattern execution.

The two forms of pattern control are:

Microcode	A microcode controls a single function, such as stamping a DCTime on or triggering a capture segment for example.
PSets	A PSet defines a programmed state for an instrument, so that a DCTime PSet contains parameters such as setting up the front end. When an instrument receives a PSet from a pattern, it enters the programmed state represented by that PSet.

The following topics are available:

- Programming Patterns in a Text Editor
- Optimizing Pattern Restart Time
- DCTime Pattern Microcodes

	• PSets
--	---

For information, see [DC Instrument Behavior During Pattern Execution](#).

[Pattern Compiler](#)

The [pattern compiler](#) takes a [text](#) .atp file and compiles it into a binary .pat file. You can edit this file using PatternTool. It is the format accepted by the pattern loader.

[Pattern Reverse Compiler](#)

The [pattern reverse compiler](#) takes a [binary](#) .pat file and compiles it into a text .atp file. This file can be edited using a text editor.

[PatternTool](#)

PatternTool allows you to read and edit binary pattern files. Using PatternTool, you can display a compiled pattern file in easily readable, column formats. You can also directly edit a pattern file, instead of having to edit the ASCII source file and recompile it. You can then update the tester with the edited pattern file.

PatternTool is useful during debugging, specifically in debugging pattern files. PatternTool can also display HRAM (History RAM) data, as captured on the tester.

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DCTime Programming : [DCTime Pattern Control](#) : Programming Patterns in a Text Editor

Programming Patterns in a Text Editor

You can create and edit a pattern using a text editor. [This form](#) (.atp file) must be compiled into a .pat file before it can be loaded. The pattern contains microcode to set instrument parameters. Use the PSet microcode to program the instrument to the parameters specified in a PSet.

[Creating a pattern in this form requires you to format the file in a specific way. A pattern requires that an instrument statement for each instrument is added and that each vector is written out in a particular way.](#)

[The following shows how to create a pattern with a text editor:](#)

- [Instruments Statement](#)
- [Vector Statements](#)
- [Pattern Example](#)

Instruments Statement

To use a DCTime from within a pattern, you must add the pin to the instruments statement, which appears at the top of the file:

```
instruments = {  
    MyPin:DCTime;  
}
```

For more information, see [Instruments Statement](#) in the Pattern Language section.

Vector Statements

Within a vector statement, a DCTime pin is programmed like other instruments that use microcodes:

```
(MyPin = Stamp) > TSet ;
```

If a pin has multiple resources behind it, then the statement appears a bit differently, with the instrument type included:

```
(MyPin:DCTime = Stamp) > TSet ;
```

Pattern Example

The following is an example of a pattern in .atp form created in a text editor for the DCTime.

```
import tset ts1;
instruments = {
    MyPin:DCTime;
}
srm_vector MyPattern ( $tset, MyDigitalPin)
{
    (MyPin:DCTime = Enable_Gate_On)
        > ts1 0 ;
        > ts1 0 ;
    (MyPin:DCTime = Stamp)
        > ts1 0 ;
    repeat 1000 >ts1 0 ;
    (MyPin:DCTime = Enable_Gate_Off)
        > ts1 0 ;
        > ts1 0 ;
    halt > ts1 0 ;
}
```

Patterns in PatternTool

The figure shows the .atp file compiled using the Pattern Compiler and loaded into PatternTool.

TSet	Command	Label	Vector/Cycle	MyPin (DCTime)	MyDigitalPin	Comment
ts1			0	Enable_Gate_On	0	
ts1			1		0	
ts1			2	Stamp	0	
ts1	repeat 1000		3		0	
ts1			4	Enable_Gate...	0	
ts1			5		0	
ts1	halt		6		0	

Pattern in PatternTool

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DCTime Programming : DCTime Pattern Control : Optimizing Pattern Restart Time

Optimizing Pattern Restart Time

IG-XL can speed up restarts of patterns that contain DC instruments. If you have run a pattern and you do not use VBT to change any parameters that affect the pattern before the next start of the same pattern, enabling optimization allows the pattern to restart more quickly. Improvement can range from 20% to 50%. The first run of a pattern always requires the full amount of time to start. After you have run a pattern once, subsequent restarts (including starts from a different label in the same pattern) benefit from the reduced startup time.

To achieve this restart efficiency, the following parameters must remain constant from one start to the next:

- Active site
- All DC PSet parameters (if you reprogram a PSet, do not apply it to hardware before you finish running the pattern)
- Source signal loaded in instrument memory

Note that faster pattern restarts may require adjustments to test program timing.

To enable pattern restart optimization, set the Related Topics property to True.

Example

The following example runs the pattern three times. The first run, starting at label t0, requires the full startup time. The second run, a restart, and the third run, a start at a different label in the pattern, are both optimized.

```
***** Start at label t0 *****
```

```
If TheExec.Sites.ActiveCount > 0 Then
```

```
TheHdw.Digital.Patterns.Pat("regulator").Start ("t0")
End If
TheHdw.Digital.Patgen.HaltWait
'***** Do something that does not change any DC instrument settings *****

'***** Restart at label t0 *****
If TheExec.Sites.ActiveCount > 0 Then
    TheHdw.Digital.Patgen.Restart
End If
TheHdw.Digital.Patgen.HaltWait
'***** Start at label t1 *****
If TheExec.Sites.ActiveCount > 0 Then
    TheHdw.Digital.Patterns.Pat("regulator").Start ("t1")
End If
TheHdw.Wait (0.01)
TheHdw.Digital.Patgen.HaltWait
```

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DCTime Programming : [DCTime Pattern Control](#) : DCTime Pattern Microcodes

DCTime Pattern Microcodes

DCTime pattern microcodes are instructions that you can place on an individual vector of a digital pattern. For that vector, a microcode instructs the DCTime to perform a particular action. A microcode controls a single function such as to trigger a new capture or make a time stamp. The format for specifying a DCTime microcode on a vector is either of the following:
(pin-list:instrument = microcode)
((pin-list1:instrument, pin-list2:instrument) = microcode)
where:

pin-list	Is the pin name or pin group or (comma-separated list of pins) as specified in the instruments statement.
instrument	Is DCTime. This corresponds to the instrument type for pin-list as defined in the instruments statement.
microcode	A DCTime microcode.

The following table lists DCTime microcodes.

DCTime Microcodes	
Microcode	Description
Enable_Gate_On	Turns on the Enable Gate window. While this Gate is open, the time stamper is enabled and each event that occurs is stamped.
Enable_Gate_Off	Turns off the Enable Gate window.

Stamp	Causes a time stamp on the time stamper set up with a pattern as the event input.
Enable_Inst_Cond	Enables the time stamper to drive the condition bit.
Disable_Inst_Cond	Disables the time stamper from driving the condition bit.
Clear_Cond	Clears the latched condition, without disabling the condition bit from being driven by events on the DCTime.
Enable_Alarm	Opens alarm window for DCTime alarms.
Disable_Alarm	Closes alarm window for DCTime alarms.
Enable_Trig	Provides an Enable Trigger signal to the Enable input of the dual stamper if the Enable input is set to listen for a microcode. Only the first event is stamped after each Enable.
Trig <name>	Starts a capture with the given name as specified in the most recent PSet. See Trig Microcode.
PSet <name>	Applies the PSet of the given name. See PSets.

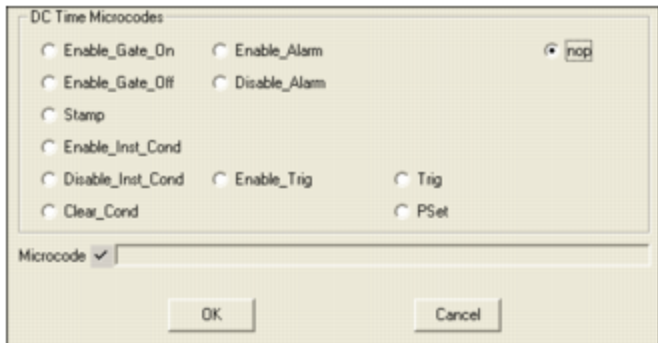
For more information, see:

- Instrument Microcode
- Pattern Example

DCTime PatternTool GUI

The PatternTool provides columns for the DCTime pins listed in the instrument statement of the compiled .atp file.

You can conveniently produce the comma-separated microcode equivalent by checking the check boxes as shown in the figure.



DCTime PatternTool Microcode Display

For more information, see:

- [Pattern Example](#)
- [About PatternTool](#)

Trig Microcode

The [DCTime](#) accepts the Trig <name> syntax pattern. You assign names to signals in Mixed Signal Workshop. IG-XL ensures that all signals used by a pattern are loaded when the pattern is loaded into pattern memory. Sixteen unique signal names can be stored in each channel’s memory. If you define and load more than 16 signals per channel per pattern, IG-XL returns an error when loading the pattern.

[For more information, see Mixed Signal Workshop.](#)

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DCTime Programming : DCTime Pattern Control : PSets

PSets

PSets control the state of an instrument for testing, such as the force voltage value and force voltage range. PSets are programmable, and the characteristics are set up using its parameters. You define a PSet in VBT language and apply it using the PSet microcode. PSet properties and methods for DCTime are listed in the table.

DCTime PSets	
DCTime PSet Property	Description
FrontEnd	Programs the front end properties and methods
Hysteresis	Programs the DCTime’s front end hysteresis.
Threshold1	Programs the DCTime’s front end threshold1.
Threshold2	Programs the DCTime’s front end threshold2.
VoltageRange	The voltage setting and the impedance of the front end.
Interleave	Programs the interleave setting of the DCTime.
Measurement	Accesses properties and methods to set up IG-XL predefined measurement types.
DutyCycle	Accesses properties and methods to set up a duty cycle measurement.
Fall Time	Accesses properties and methods to set up a fall time measurement.

Frequency		Accesses properties and methods to set up a frequency measurement.
Period		Accesses properties and methods to set up a period measurement.
Pin to Pin Delay		Accesses properties and methods to set up a pin-to-pin delay measurement.
Pulse Width		Accesses properties and methods to set up a pulse width measurement.
Rise Time		Accesses properties and methods to set up a rise time measurement.
Stamper		Specifies either stamper1 or stamper2 properties.
Enable		Programs the stamper's enable properties and methods.
	GetInput	Gets the input for the DCTime enable pin.
	Mode	Gets or sets the mode for the stamper.
	SetInput	Sets the input for the DCTime start pin.
	Slope	Gets or sets the slope for a measurement enable signal.
Event		Programs the stamper's event properties and methods.
	Input	The event that will trigger the time stamper for the stamper.
	Slope	The event slope for the stamper.
HoldOff		Programs the hold off properties and methods for the specified stamper.
	Count	The number of events to count during hold-off for the PSet's first stamper.
	Mode	The mode of hold-off for the stamper.
	Time	The amount of time to wait during hold-off for the stamper.
SampleSize		The number of time stamps to be taken by the stamper.
Start		Specifies the start interface including slope information.
	GetInput	Gets the input for the DCTime start pin.
	SetInput	Sets the input for the DCTime start pin.

Slope

The slope of the start signal.

PSet Programming Example

You define a PSet in VBT whether you are creating patterns with PatternTool or a text editor. The following example shows all PSet programmable parameters being programmed for the DCTime. This PSet can then be applied from a pattern whenever these parameters are needed.

' Create a PSet and add it to the PSet collection
TheHdw.DCTime.Pins("MyPin").PSets.Add "MyPSet"
' Program the hardware parameters to the desired values

With TheHdw.DCTime.Pins.PSets.Item("MyPSet")
 .FrontEnd.Hysteresis = tIDCTimeHysteresisOff
 .FrontEnd.Threshold1 = 2
 .FrontEnd.Threshold2 = 4
 .Interleave = tIDCTimeInterleaveOff

End With

' Load the PSet into PSet memory
TheHdw.DCTime.pins("MyPin").PSets("MyPSet").Apply

Applying a PSet

You apply a PSet from either a pattern or to the instrument in VBT. To apply a PSet with VBT use the following syntax:

TheHdw.DCTime.Pins("MyPin").PSets.Item("MyPSet1").Apply

A PSet called from a text file would use the following format:

(MyPin:DCTime = PSet MyPSet)

To call a PSet in PatternTool the setup would look as follows:

TSet	Command	Label	Vector/Cycle	MyPin (DCTime)	digitalpin1	Comment
ts1			0		0	
ts1			1	PSet MyPSet	0	
ts1			2		0	

PSet Called Using PatternTool

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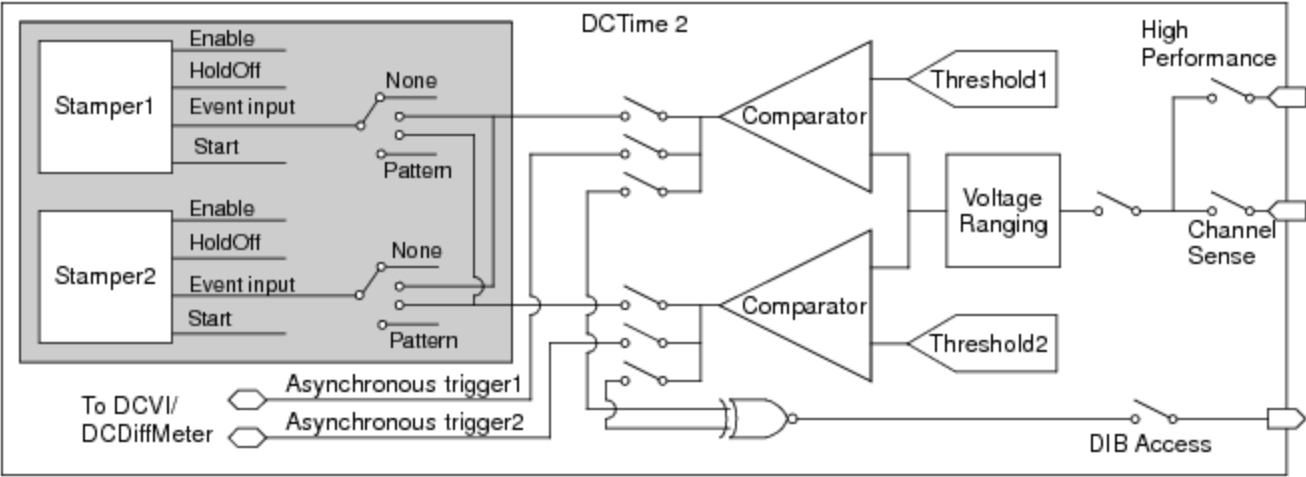
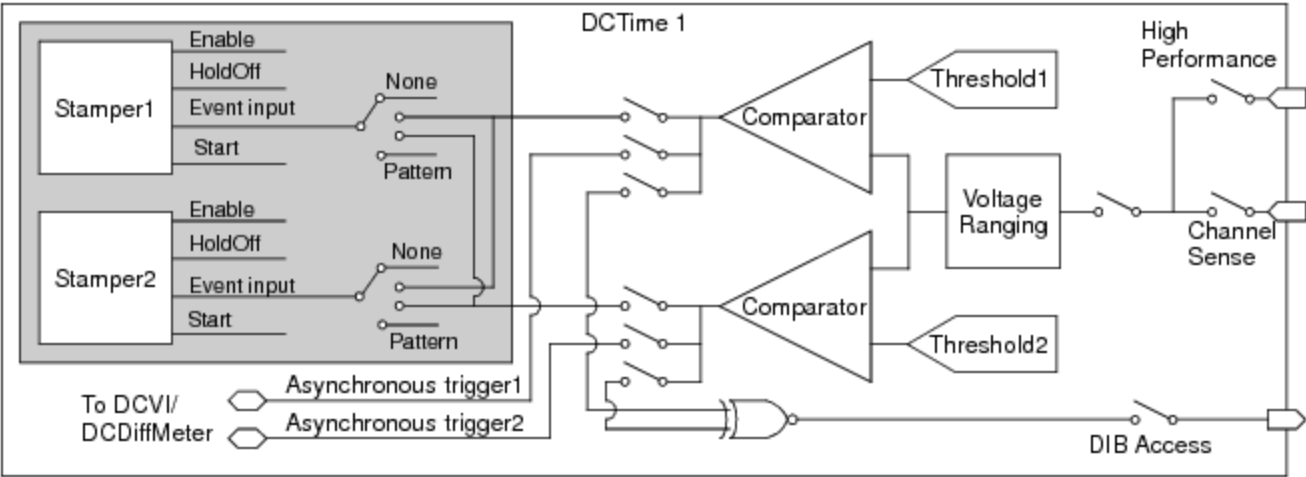
DCTime Theory of Operation : DCTime Overview

DCTime Overview

A DC Time Measurement Unit connects to the DUT using the channel sense lines. Depending on the instrument board, the DCTime connects to a different number of channels. The following table shows the sense channels a DCTime connects to on each instrument board:

DCTime Channel Access		
DCTime	DC30 Channels	DC75 Channels
1	1, 2, 5, 6, 9, 10, 13, 14, 17, 18	1, 2
2	3, 4, 7, 8, 11, 12, 15, 16, 19, 20	3, 4

Each DCTime instrument is comprised of a selectable front end voltage range, two programmable threshold comparators, and two time stampers. Signals can be connected to the front end through preassigned channel sense lines or through a dedicated high performance input. The following figure is a block diagram of the DCTime.



Block Diagram of Two DCTime Instruments

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DCTime Theory of Operation : [DCTime Overview](#) : Voltage Range and Impedance

Voltage Range and Impedance

Each DCTime interfaces to the DUT through a selectable front end that attenuates and/or buffers the input signal. You can select from among several high impedance voltage ranges or a 10V 50ohm range as required by the application.

The DCTime has the following voltage/impedance ranges:

DCTime Voltage Range/Impedance Ranges	
DC30	DC75
10V, HiZ	20V, HiZ
30V, HiZ	75V, HiZ
10V, 50ohms	10V, 50ohms

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DCTime Theory of Operation : [DCTime Overview](#) : Comparators

Comparators

A comparator compares the attenuated input signal and the programmed voltage threshold level. The event detection slope is programmable to positive or negative for each stamper.

Each comparator has selectable hysteresis feature. See [Hysteresis](#).

Each of the threshold comparators has an independent trigger level DAC and programmable hysteresis. The threshold comparators generate an edge each time the input signal transitions past the programmed threshold level. The output of each comparator is routed to a time stamper capable of performing several types of time measurements. Additionally, the comparator outputs can be routed to the internal asynchronous trigger bus to initiate additional DCVI or DCDiffMeter measurements as well as to the DIB.

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DCTime Theory of Operation : [DCTime Overview](#) : Stamper

Stamper

Each DCTime has a programmable dual time stamper to use to detect a comparator edge transition to take a measurement, enable measurements, or count events to hold off before taking a measurement. Additional programming parameters necessary to configure a stamper are the number of samples to stamp, a measurement start command, and the slope of the comparator edge to detect. You can program all these parameters using VBT or pattern PSets and microcodes. For more information, see:

- | | |
|---|-------------------------------|
| • | Pins.Stamper1 |
| • | PSets |

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DCTime Theory of Operation : [DCTime Overview](#) : [DCTime Counter](#)

DCTime Counter

The DCTime has one counter for the dual stamper, which means that two stampers share one counter. The DCTime counter is circular.

A circular counter is a continuous running counter that wraps over at a certain point rather than maintaining a running total of time. One full rotation through the DCTime counter is 671.09ms. On reaching this point, the counter resets to zero and repeats the rotation. The counter then reports the value at which it has stopped on the counter (0 to 671.09ms), not the cumulative time. The figure shows the counter as it cycles through time with the elapsed time increasing until it reaches 671.09ms where it resets to 0.



DCTime Counter Wrapping Example

The hardware does not track if the counter has wrapped or how many times it has wrapped.

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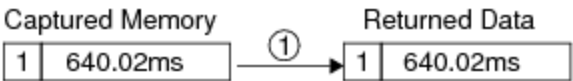
DCTime Theory of Operation : DCTime Overview : Counter Example

Counter Example

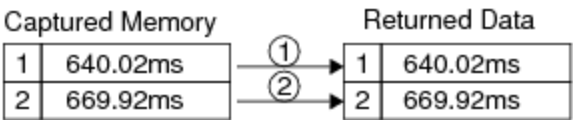
A DCTime can make different types of measurements, such as rise or fall time measurements. Each measurement type requires an event to activate the stampers to make a time stamp. When a stamp occurs, the DCTime stamps the current counter time into capture memory. Since the DCTime counter is circular, it must be determined if the counter wrapped during stamping. The hardware does not track if the counter has wrapped, but IG-XL unwraps each stamper’s data separately. IG-XL does not track wrapping between two stampers unless you use the .CalculatedResults method.

When a captured result is lower than the previous captured result, IG-XL assumes that the counter has wrapped and returns data based on this. The following sequence shows how this calculation is done:

1. The time stamped to memory is returned as it was captured. Wrapping is not assumed here since this is the first event stamped.



2. The time stamped to memory is returned as it was captured. Wrapping is not assumed here since this value is higher than the previous value. If the sample rate is larger than 671.09ms you must track the possibility that the counter wrapped.



✓ Note:

If the time interval between events is longer than the maximum counter value of 671.09ms, the counter can wrap and have a time stamp larger than the previous number. In this case, you must track wrapping based on the predicted sample rate.

- The time stamped to memory is lower than the previous value, and IG-XL assumes that the counter has wrapped. Since the counter has wrapped, 671.09ms is added to the stamped value to acquire the returned value.

Captured Memory			Returned Data	
1	640.02ms	①	1	640.02ms
2	669.92ms	②	2	669.92ms
3	027.37ms	③	3	698.46ms

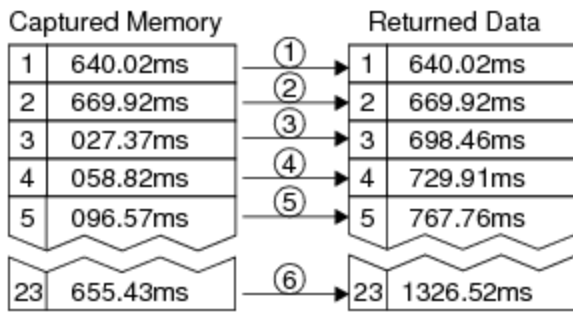
- Captured memory is more than the previous sample, but IG-XL has already noted that the counter wrapped once. Thus, 671.09ms is added to captured memory.

Captured Memory			Returned Data	
1	640.02ms	①	1	640.02ms
2	669.92ms	②	2	669.92ms
3	027.37ms	③	3	698.46ms
4	058.82ms	④	4	729.91ms

- Again, captured memory is more than previous sample, but IG-XL has already noted that the counter wrapped once. Therefore, 671.09ms is added to captured memory.

Captured Memory			Returned Data	
1	640.02ms	①	1	640.02ms
2	669.92ms	②	2	669.92ms
3	027.37ms	③	3	698.46ms
4	058.82ms	④	4	729.91ms
5	096.57ms	⑤	5	767.76ms

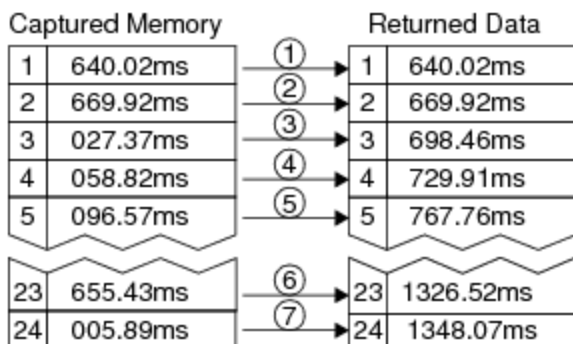
- Again, captured memory is more than previous sample, but IG-XL assumes that the counter wrapped once. Therefore, 671.09ms is added to captured memory.



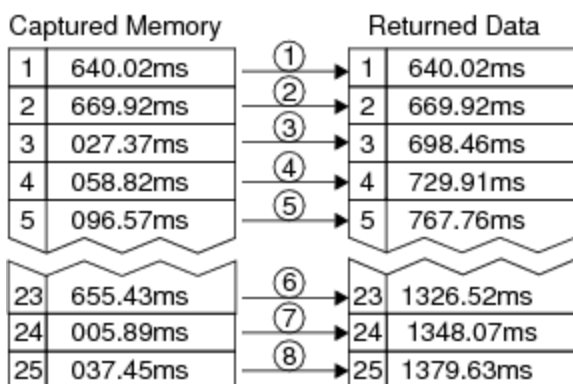
7. The time stamped to memory is lower than the previous value, IG-XL assumes that the counter has wrapped twice. Since the counter wrapped twice, 671.09ms is added twice to the stamped value to acquire the returned value.

✓ Note:

The DCTime cannot determine if the counter has wrapped more than once. For events longer than twice the counter length you must track wrapping based on the expected time interval between events. Adjust the returned value by adding $N * 671.09\text{ms}$ to the measurement result where N represents the number of times the counter is expected to wrap greater than 1.



8. Captured memory is more than previous sample, but IG-XL has noted that the counter has wrapped twice. Therefore, 671.09ms times two is added to captured memory.



For more information, see:

•	Measurement Types
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•	CalculatedResults Method
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[DCTime Theory of Operation](#) : [DCTime Overview](#) : DCTime High Performance Input

DCTime High Performance Input

Each DCTime has a dedicated connection to the DUT through a high performance input line. This 50ohm line is a higher bandwidth path than the DCVI sense lines, as it bypasses the capacitance of the DCVI channel path. You use this connection by declaring the pin as DCTimeHP in the channel map and selecting the desired .dig channel. Each DCTime has a corresponding .dig channel (.dig1 -> DCTime1, .dig2 -> DCTime2).

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[DCTime Theory of Operation](#) : [DCTime Overview](#) : [DCTime Guard](#)

DCTime Guard

When connecting the DCTime through a DC channel, the DCTime grounds the DCVI guard pin to improve signal integrity. Note that since the DCVI uses the guard pin as a driven guard it must be disconnected from the DIB before using that channel to make a time measurement.

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[DCTime Theory of Operation](#) : [DCTime Overview](#) : [DIB Connection for Digital Edge Finds](#)

DIB Connection for Digital Edge Finds

The output of the DCTime front ends can be connected to the DIB to perform digital edge-find measurements (using a digital channel) or trigger other measurements at the DIB. This feature is useful for making propagation delay measurements between a digital pin and an analog pin. This option is also useful for making measurements that require improved accuracy over the local time stampers.

The two comparators on each DCTime can connect to the DIB through an XOR gate. This provides a pulse at the DIB where each edge indicates when an event was detected.

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DCTime Theory of Operation : Measurement Types

Measurement Types

The DCTime can make measurements using predefined measurement methods, such as period and frequency. Using a predefined method, IG-XL programs the instrument for the selected measurement type, requiring you to program only a subset of parameters. However, not all functionality of the DCTime is available. For example, holdoff and enable are both off.

The following topics are available:

- [Period Measurements](#)
- [Frequency Measurements](#)
- [Pulse Width Measurements](#)
- [Duty Cycle Measurements](#)
- [Rise or Fall Time Measurements](#)
- [Pin-to-Pin Delay Measurements](#)

Event Capture Programming

You can program every parameter independently for greater control of the instrument using event capture programming. Event capture programming gives you full control of all functionality of the DCTime, including such options as enable and holdoff. However, you are required to manage the time stamps and the setup of the instrument.

Results are returned in VBT as two PinListData objects that must be calculated to determine the results.

For more information, see Pins.Stamper1.Event in the VBT Syntax Reference.

CalculatedResults Method

The starting stamp must be calculated for all measurements that use two stampers. Using the .CalculatedResults method in VBT, you can return a measurement using syntax such as the following:

```
pldResults = TheHdw.DCTime.Pins("MyPin").Measurement.Period _  
    .CalculatedResults(pld1, pld2)
```

What occurs is that each time stamper has a time value. Since the DCTime counter wraps, a stamper with a larger number may have gone first.

If the time from stamper 1 is larger than the time from stamper 2, then the following formula is used to determine which stamper stamped first:

If $TS1[0] - TS2[0] > TS2[0] - TS1[0] + \text{maximum time}$

TS1 was first

Else

TS2 was first

If the time from stamper 2 is larger than the time from stamper 1, the following formula is used to determine which stamper stamped first:

If $TS2[0] - TS1[0] > TS1[0] - TS2[0] + \text{maximum time}$

TS2 was first

Else

TS1 was first

The DCTime can only correctly determine which stamper has stamped first if the time difference between the two stampers is less than the maximum time (670ms) divided by two.

If the delay is greater than the maximum counter time, you must calculate measurements manually.

For more information, see:

- [DCTime Counter](#)

- [Counter Example](#)



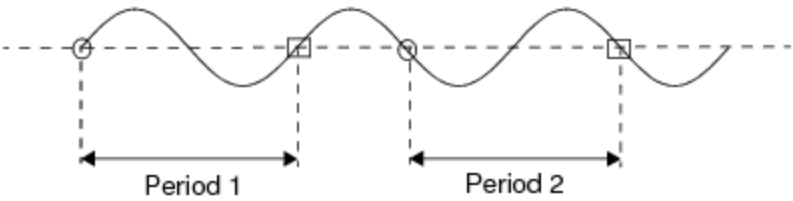
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DCTime Theory of Operation : [Measurement Types](#) : Period Measurements

Period Measurements

Computes the time from the detection of an event on the input signal until a second detection of the same event on the same signal. In the figure, stamper 1 stamps the beginning and stamper 2 stamps the finish. The time between the beginning and the finish is the period. The finish stamp for one period cannot be used as the start stamp for a second period measurement. A new beginning stamp must occur.



Period Measurement Example

For more information, see:

- [Pins.Measurement.Period](#) in the VBT Syntax Reference
- [Period Measurement](#)

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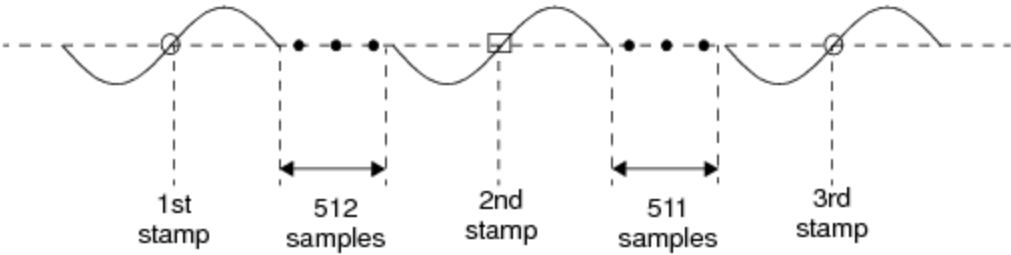
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DCTime Theory of Operation : Measurement Types : Frequency Measurements

Frequency Measurements

Measures frequency over user specified number of periods of an input signal. Only one stamper is used for a frequency test. The signal must have a period time that is less than 671ms, which means that the frequency must be greater than 1.49Hz.

You can specify number of periods over which to take the measurement from 1 - 512 or any integer multiple of 512 up to 262,144. For example, if 526 periods are requested, it is assumed the next highest integer multiple of 512 is desired, so the actual number of periods over which the measurement is taken is 1024. The figure shows a frequency measurement with number of periods over which the measurement is taken at 1024.



Frequency Measurement with Number of Periods 1024

For more information, see:

- [Pins.Measurement.Frequency](#) in the VBT Syntax Reference
- [Frequency Measurement](#)

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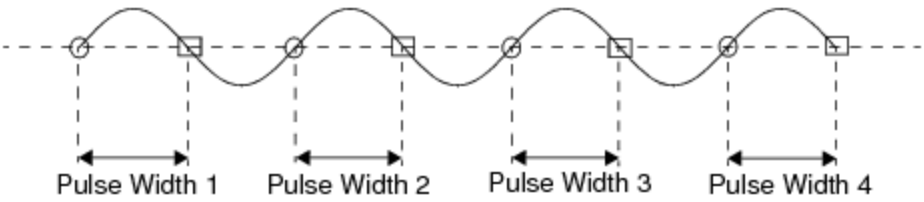
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DCTime Theory of Operation : [Measurement Types](#) : Pulse Width Measurements

Pulse Width Measurements

Computes the time from an event on the input signal to another event on the input signal with the identical threshold voltage and the opposite slope. The figure shows stamper 1 stamping the beginning on the rising (positive) slope and stamper 2 stamping the finish on the falling (negative) slope. The pulse width is the difference between the two.



Pulse Width Measurement Example

For more information, see:

- [Pins.Measurement.PulseWidth](#) in the VBT Syntax Reference
- [Pulse Width and Duty Cycle Measurements with Hysteresis](#)
- [Pulse Width](#)

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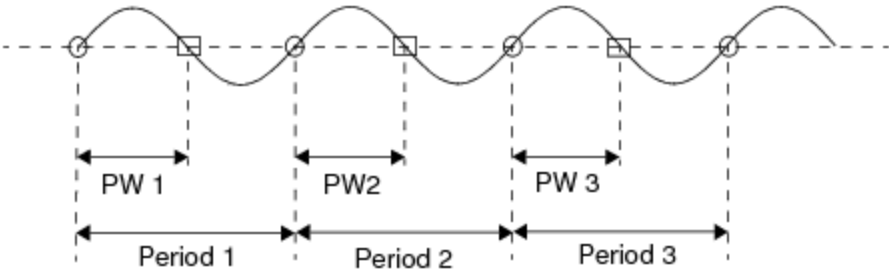
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DCTime Theory of Operation : Measurement Types : Duty Cycle Measurements

Duty Cycle Measurements

Measures the ratio of input signal pulse width to period. For the pulse width, stamper 1 stamps the beginning and stamper 2 stamps the finish. For the period measurement, stamper 1 stamps both the beginning and finish. The figure shows an example of the stamps each stamper makes.



Duty Cycle Measurement Example

For more information, see:

- [Pins.Measurement.Period](#) in the VBT Syntax Reference
- [Pulse Width and Duty Cycle Measurements with Hysteresis](#)
- [Duty Cycle Measurements](#)

Pulse Width and Duty Cycle Measurements with Hysteresis

When using the predefined pulse width or duty cycle measurement methods, the DCTime uses a single threshold setting to achieve the best trigger level accuracy, which means only one hysteresis setting can be programmed.

For more information, see [Hysteresis](#).

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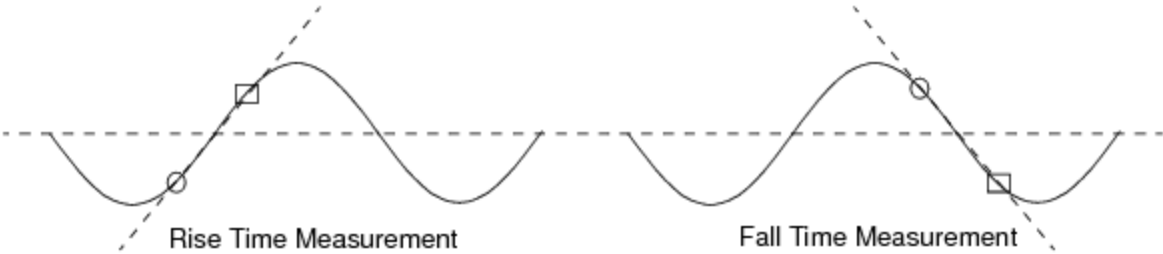
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DCTime Theory of Operation : Measurement Types : Rise or Fall Time Measurements

Rise or Fall Time Measurements

Measures the time an input signal takes to travel between two thresholds with the same slope. The figure shows that stamper 1 stamps the beginning and stamper 2 stamps the finish. Both a rise and a fall time measurement are shown.



Rise and Fall Time Measurement Example

For more information, see:

- [Pins.Measurement.FallTime](#) in the VBT Syntax Reference
- [Pins.Measurement.RiseTime](#) in the VBT Syntax Reference
- [Rise Time Measurement](#)

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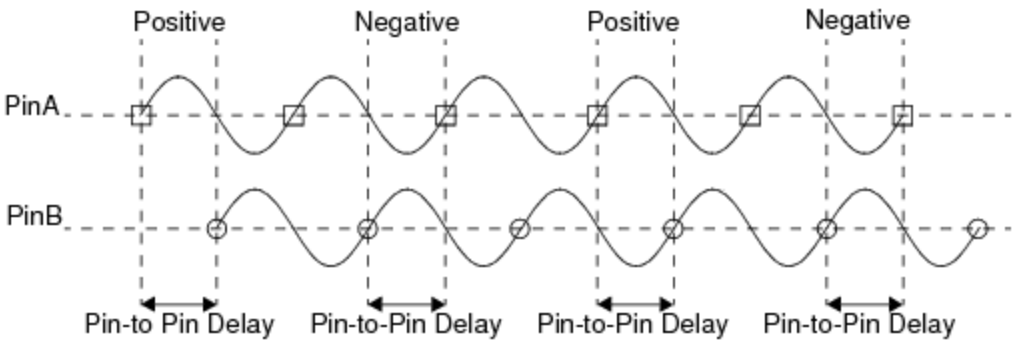
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DCTime Theory of Operation : Measurement Types : Pin-to-Pin Delay Measurements

Pin-to-Pin Delay Measurements

Measures the time between a specified event on one input and a separately specified event on another input. Each input must be connected to a separate DCTime on the same DC instrument board. The figure shows PinA connected to stamper 1 of a DCTime and PinB connected to stamper 1 of a second DCTime on the same DC instrument board. The difference between the stamp on PinA and stamp on PinB is the pin-to-pin delay.



Pin-to-Pin Delay Measurement Example

If the result of a pin-to-pin delay is negative, the event on PinB occurred before PinA. For more information, see:

- [Pins.Measurement.PinToPinDelay](#) in the VBT Syntax Reference
- [Pin-to-Pin Propagation Delay Measurements](#)

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DCTime Theory of Operation : Start Modes

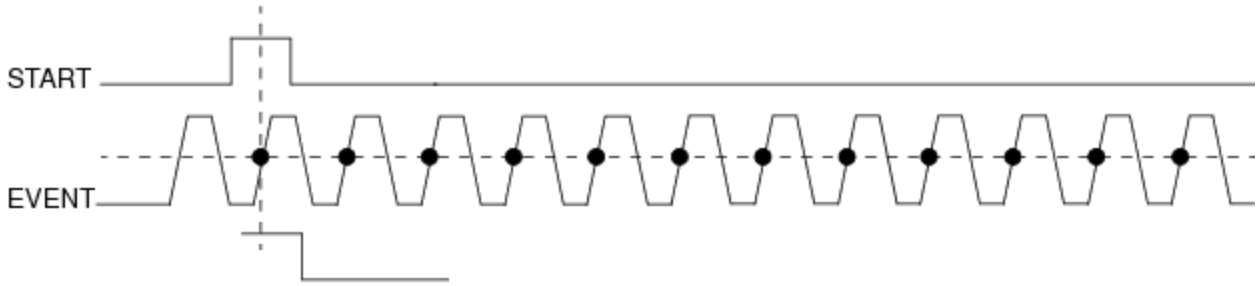
Start Modes

When a start signal is received, the DCTime makes measurements until it finishes the programmed capture or until a stop command is issued. The start signal is made only for a single specified measurement. When the start is executed, the program determines if the trigger signal has occurred and if the stamper events is programmed to something other than none. If the trigger has not occurred then the DCTime does nothing.

The DCTime has two start modes:

OnTrigger	The measurement automatically begins with the first event detected after the start command is generated in VBT or a pattern.
Pin	The time stamper is started by an another DCTime on the same DC instrument board after the start command is generated in VBT or a pattern.

DCTime has one start signal, which is shared by both stampers. The measurement begins on the next event after the start event. The figure shows a start signal from an external event and captures on each event after the start. This example assumes that enable input is programmed to none and that hold-off mode is off.



Starting DCTime in Continuous Mode Using an External Event

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DCTime Theory of Operation : Enable Mode

Enable Mode

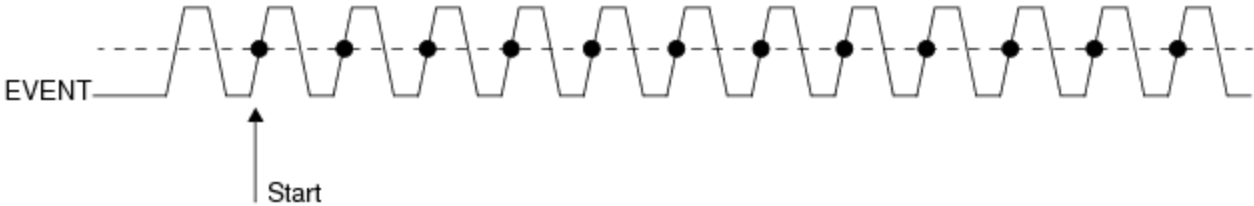
Enable allows you to control which edges are captured. After the start signal, the stamper is ready to stamp and is just waiting for an enable signal.

Enable Mode controls when each individual stamper can stamp. DCTime has three modes for selectively capturing events:

Continuous (off)	Once started, the stamper captures all subsequent events.
Trigger Each Event	Stamper must receive a trigger on its enable input before capturing the next event signal.
Gate Window	Stamper captures events as long as its enable input is at gate_high. The stamper cannot stamp if enable input is gate_low.

Continuous Mode

The figure shows the edges captured on a signal when the enable mode is programmed to Continuous. Each event that occurs is stamped.

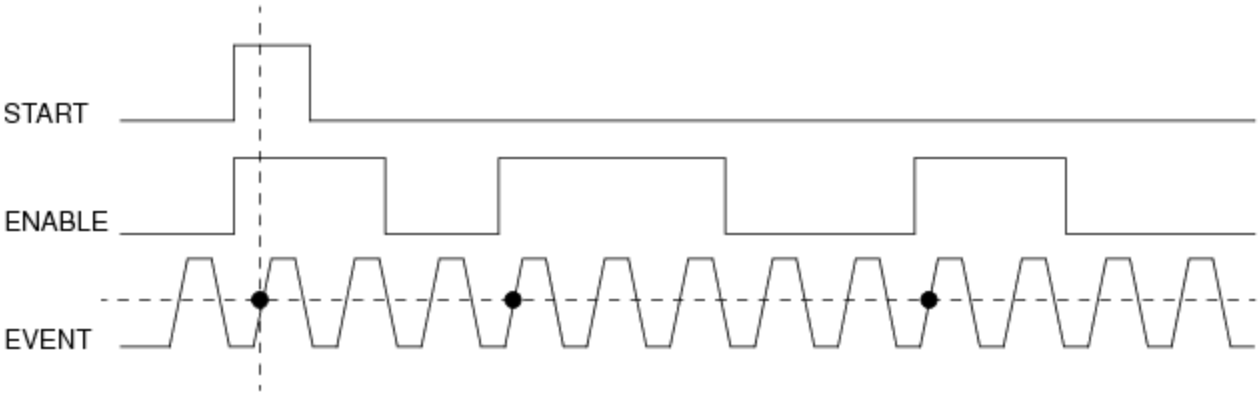


Continuous Mode

Trigger Each Event

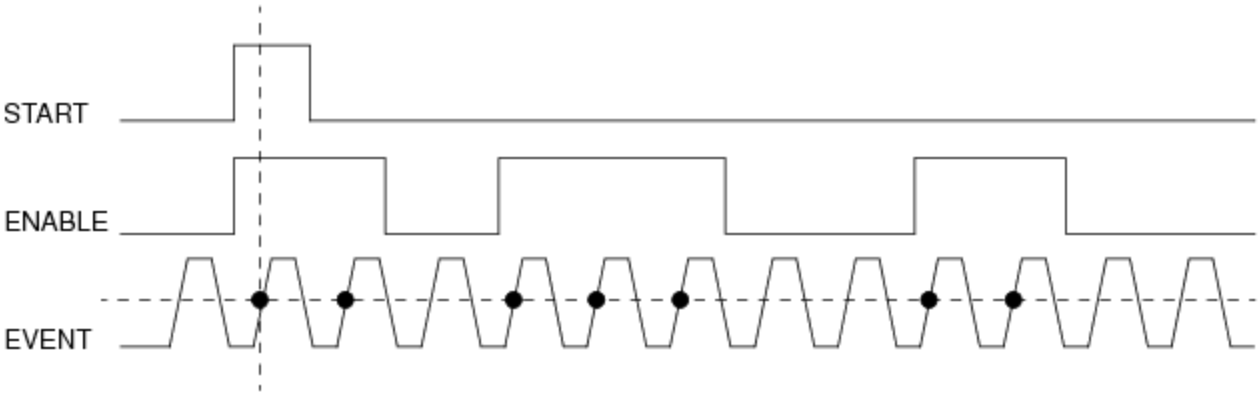
Trigger Each Event enables capture of a single event on the enable signal's set edge. The edge choice can be set by a positive or negative slope.

The figure shows the enable signal and the event that is stamped. Even though the enable signal is active for more than one event, only the first event after the enable is captured using Trigger Each Event.



Trigger Each Event Mode
Gate Window

Gate Window enables capture of one or more events by setting the level of a gate signal. After the start signal, the stamper is ready to stamp and is waiting for an enable signal. The figure shows the enable signal and the events stamped. While the stamper is enabled in Gate Window, each event that occurs is stamped.



Example of Gating the Instrument With Enable On

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DCTime Theory of Operation : Enable Mode : Event Signal

Event Signal

The event is the signal that the DCTime records time on. It is this signal that the stampers stamp a time on. The event signal has four settings:

None	The event signal is off and not used. When the event signal is set to None, do not use interleave or the hardware will wait forever.
Pattern	The event signal comes from a pattern.
Threshold 1	The event occurs when the programmed threshold 1 is reached.
Threshold 2	The event occurs when the programmed threshold 2 is reached.

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DCTime Theory of Operation : Enable Mode : Hold-Off Mode

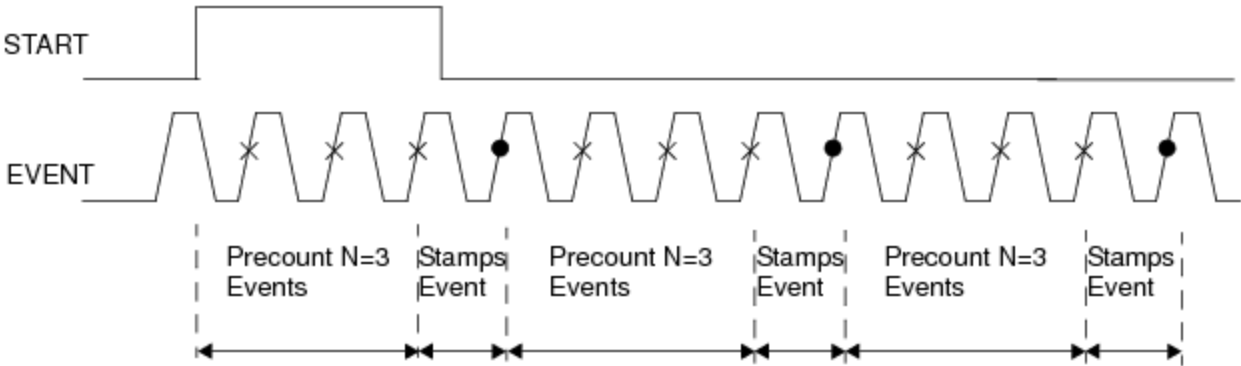
Hold-Off Mode

Hold-off mode is used to create a delay before a stamp is made. Hold-off can be set to the following:

Precount once	Counts a programmed amount of time or counts a programmed number of events before taking the measurement.
Before each	Counts a programmed amount of time or counts a programmed number of events before making each time stamp.
Off	No delay.

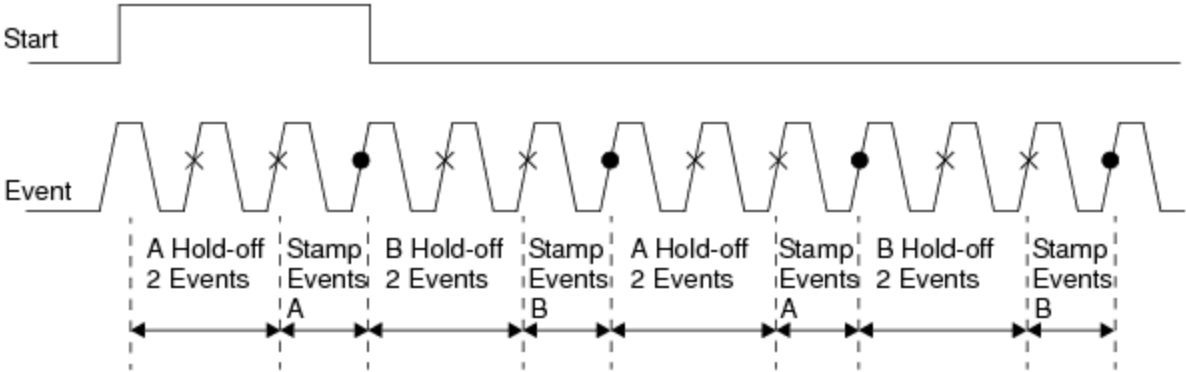
Hold-Off Count

The hold-off counters allow you to skip a specified number of events before stamping an event. For example, if you are only interested in capturing every fourth edge, you would set the hold-off counter to 3 and only capture the fourth edge. The figure shows a stamper using precounting between events to hold off three events.



Hold-Off Count Before Each Stamp Using Continuous Enable Mode

You can also use the hold-off counters as pre- and post-counters using both stampers. Here, both stampers' hold-off counters are used and programmed to count events respectively. The stampers are used in Interleave mode (see Interleave Mode) when using pre- and post-counting. The figure shows stamper1 holding off two events before stamping the next event, then stamper2 waiting two events before stamping an event. Stampers hold-off events are programmed individually and can differ from one another.



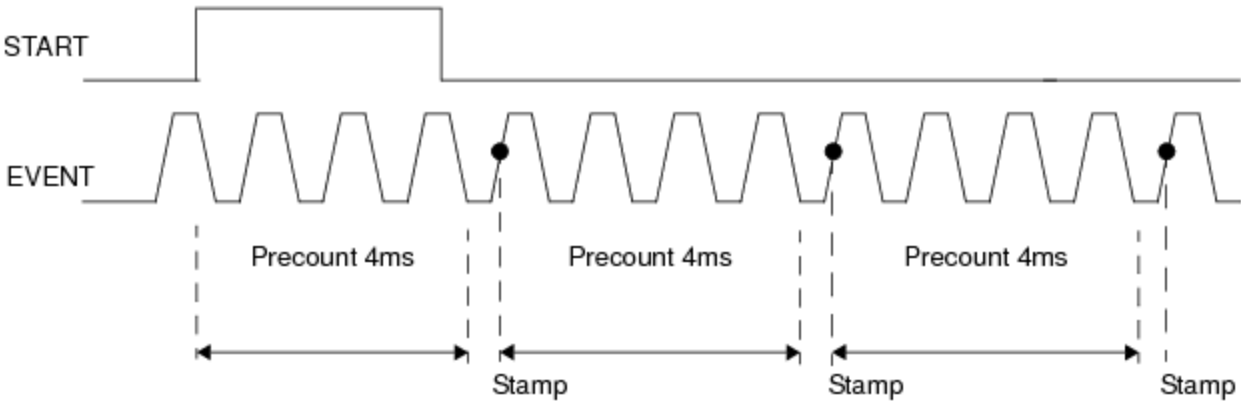
Hold-Off Count with Pre- and Post-Counting

Hold-Off Time

The hold-off counters allow you to skip a specified amount of time before stamping an event. The specified time is the minimum time the stamper waits for the next event to stamp. You can set the time in two modes:

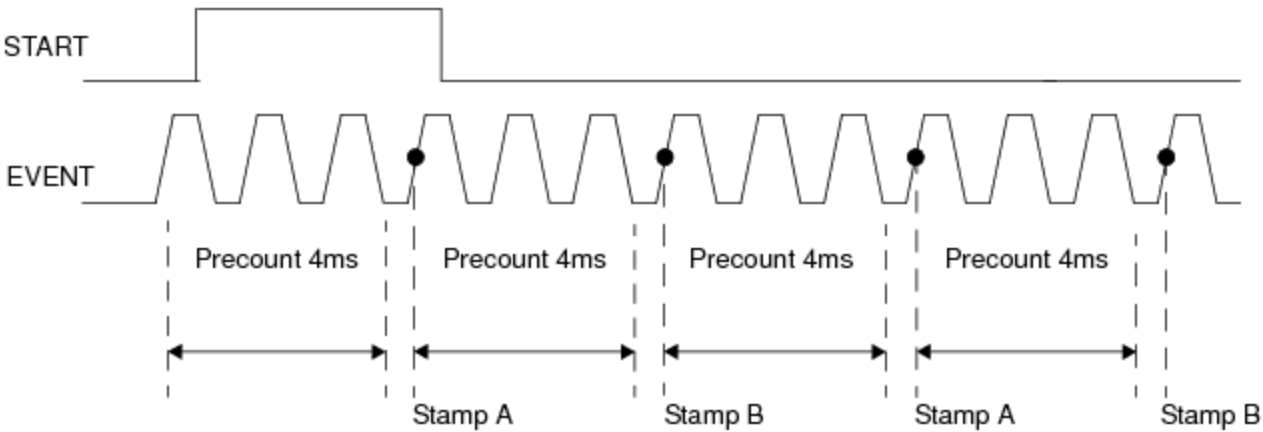
- Up to 50s at 100ns intervals
- Up to 4ms at 8s intervals

The figure shows a stamper programmed with hold-off time. The hold-off is for the programmed time, and the stamp is made on the next event. A stamp does not happen directly after the hold-off time elapses. It holds off for the programmed time plus any time that passes waiting for the next event.



Hold-Off Time in Precount Before Mode

The hold-off counters can also be used as pre- and post-counters using both stampers. Here, hold-off is used for both stampers to hold off time. The stampers are used in Interleave mode (see [Interleave Mode](#)) when using pre- and post-counting. The figure shows stamper1 holding off 4ms before stamping the first event, then stamper2 waiting 4ms before stamping an event. Stamper hold-off times are programmed individually and can differ from one another.



Hold-Off Time with Pre- and Post-Counting

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DCTime Theory of Operation : Interleave Mode

Interleave Mode

Interleave mode is useful in avoiding misalignment of time stamps. Stamper2 only captures events after Stamper1 has captured, then Stamper1 can only capture events after Stamper2 has captured.

☒ **Note:**

If either stamper is programmed to none, do not use Interleave mode or the hardware waits indefinitely.

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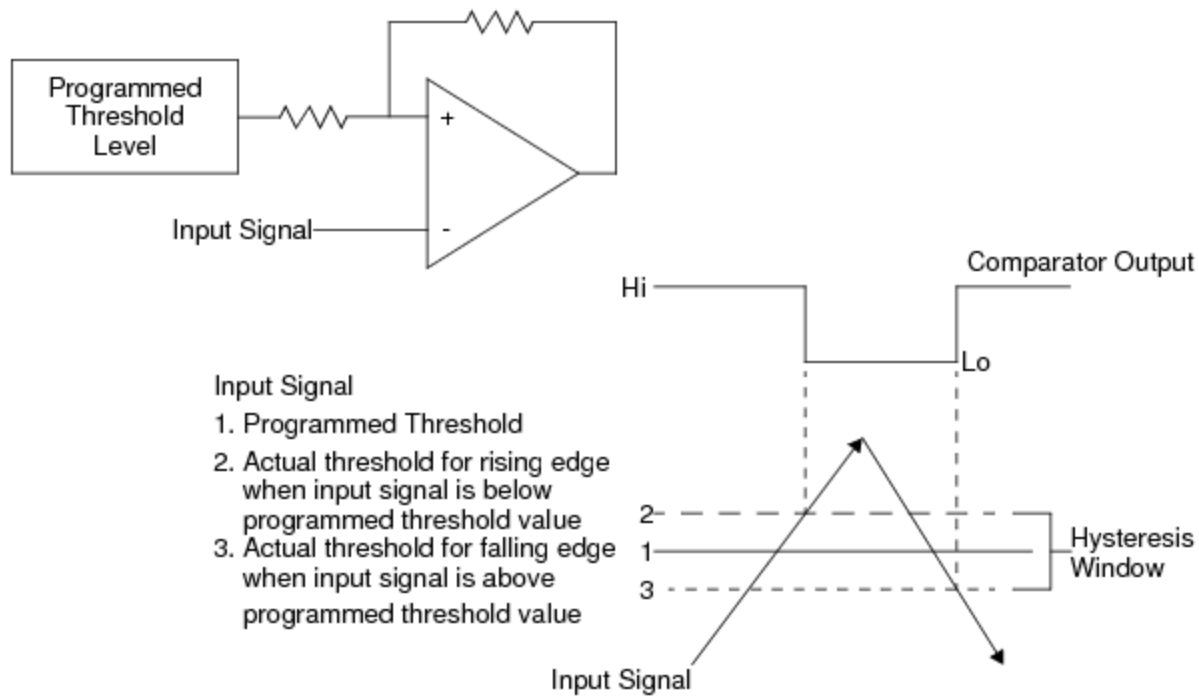
DCTime Theory of Operation : Hysteresis

Hysteresis

High speed comparator circuits can chatter, changing states multiple times as the threshold is crossed. This chatter is caused by noise from the input signal and the comparator reference. The slower the input signal crosses the threshold the more chatter that can occur.

Hysteresis corrects this undesired behavior. It changes the threshold level as the input crosses it, in such a way that once the comparator changes state it remains in the new state. With the comparator in a new state, the input signal is farther away from the new threshold level, reducing the chance of chatter.

On a signal that is not symmetric in rise and fall time within the hysteresis window for the DCTime voltage range, unexpected results with the hysteresis setting can occur. To use different hysteresis settings, you must use the DCTime in the Event Capture mode where the two thresholds can be set independently rather than using predefined measurement methods for pulse width or duty cycle measurements.



Hysteresis

The DCTime can apply the following hysteresis settings to the front end:

- Hysteresis Off
- Hysteresis On
- Hysteresis On Positive Adjust
- Hysteresis On Negative Adjust

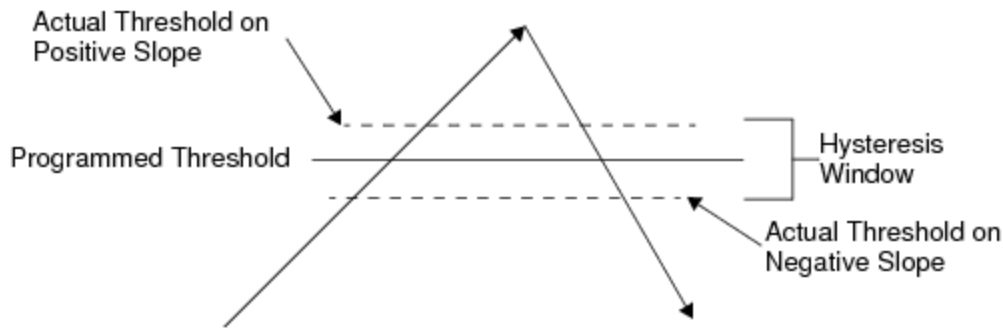
Hysteresis Off

This setting is typically used when measuring digital signals or signals with fast edges. Hysteresis off provides no noise immunity when measuring signals with slow slew rates.

Hysteresis On

This setting is typically used when performing single threshold measurements such as duty cycle, period, frequency, and pulse width of signals with symmetric rising and falling edges.

When the input signal is below the programmed threshold, the trip point is the programmed level +1/2 the hysteresis window. When the input signal is above the programmed threshold, the trip point is the programmed level -1/2 of the hysteresis window.



Hysteresis On

For example, 4.6V is programmed for the threshold level and the input signal is below the programmed trip level. The comparator changes state when the input signal crosses $4.6V + 1/2$ the hysteresis window, not at 4.6V. If the input signal is above the programmed trip level, the comparator changes state when the input signal crosses $4.6V - 1/2$ the hysteresis window, not at 4.6V.

☑ **Note:**

The maximum input signal amplitude must be greater than the threshold programmed plus one half of the hysteresis window and the minimum input signal amplitude must be less than the threshold programmed minus one half of the hysteresis window. See DC30 Specifications or DC75 Specifications.

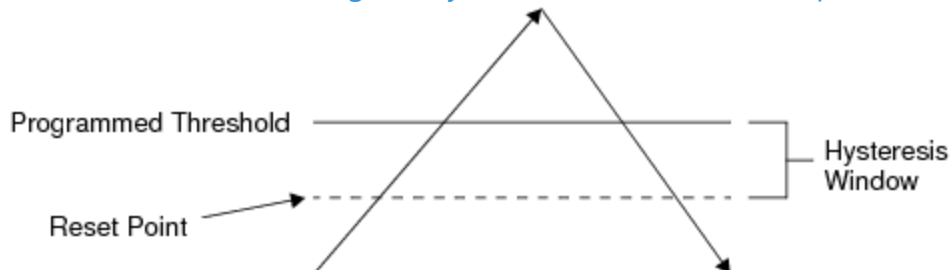
Hysteresis On Positive Adjust

This setting is typically used when performing dual threshold measurements of signals with a positive slope, such as a rise time measurement.

This setting applies the hysteresis but optimizes the levels such that for input signals with a positive slope, the actual comparator threshold is optimized to set the trip point at the DC level programmed.

For example, if 4.6V is programmed for the threshold level and the input signal is a positive adjust, then the comparator will trip at 4.6V.

This figure shows the programmed threshold, which is the level the comparator is set to change state. On the state change, the comparator does not change state again until the signal has crossed the reset point which is the hysteresis window. If the signal dropped below the programmed threshold before crossing the hysteresis window, the comparator state would not change.



Hysteresis On Positive Slope

☑ **Note:**

The level required to reset the comparator when the input signal is falling is one full hysteresis window below the programmed level. The input signal must transition below this level if multiple stamps are to be made on the rising input signal. See [DC30 Specifications](#) or [DC75 Specifications](#).

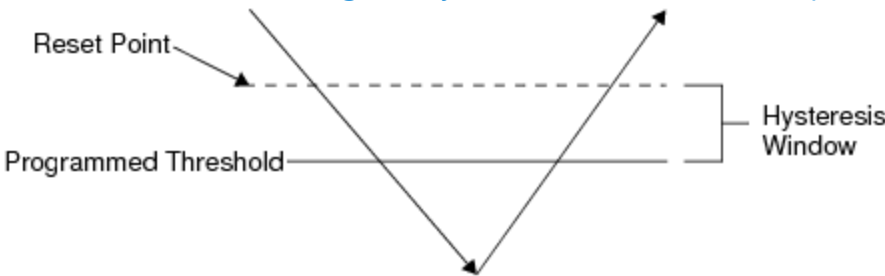
Hysteresis On Negative Adjust

This setting is typically used when performing dual threshold measurements of signals with a negative slope, such as a fall time measurement.

This setting applies the hysteresis but optimizes the levels such that for input signals with a negative slope, the actual comparator threshold is optimized to set the trip point at the DC level programmed.

For example, if 4.6V is programmed for the threshold level and the input signal is a negative adjust, then the comparator will trip at 4.6V.

This figure shows the programmed threshold, which is the level the comparator is set to change state. On the state change, the comparator does not change state again until the signal has crossed the reset point which is the hysteresis window. If the signal dropped below the programmed threshold before crossing the hysteresis window, the comparator state would not change.



Hysteresis On Negative Slope

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DCTime Theory of Operation : Asynchronous Trigger

Asynchronous Trigger

A DCTime can detect DUT state changes and asynchronously trigger meter measurements on the same DC instrument board using OnBoard trigger mode or can trigger events on other instrument boards, DIB or DUT using DIB Trigger mode.

Making an asynchronous trigger measurement requires the DCTime front end comparators. The front end comparator outputs are routed to the internal board trigger lines, which can be connected to the specified meter or the specified DIB XOR outputs. The two types of asynchronous triggering are:

- | | |
|---|--------------------|
| • | OnBoard Triggering |
|---|--------------------|
- | | |
|---|----------------|
| • | DIB Triggering |
|---|----------------|

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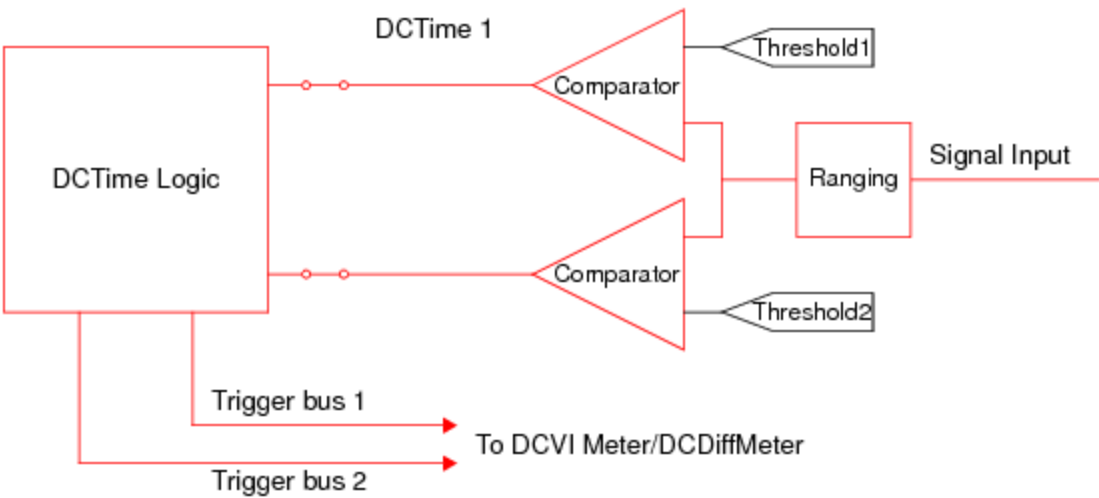
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DCTime Theory of Operation : Asynchronous Trigger : OnBoard Triggering

OnBoard Triggering

The DCTime can asynchronously trigger a DCVI or DCDiffMeter measurement on the same instrument board. Each DCTime has two trigger lines, one from each comparator that can be used to trigger the meter.

The DUT signal connects to the DCTime front end where the threshold level and voltage range can be programmed. When the threshold level is crossed a signal is output from the comparator on the trigger line to the DCVI or DCDiffMeter channel specified by the program. The figure shows the trigger line connections.



Onboard Asynchronous Trigger Lines

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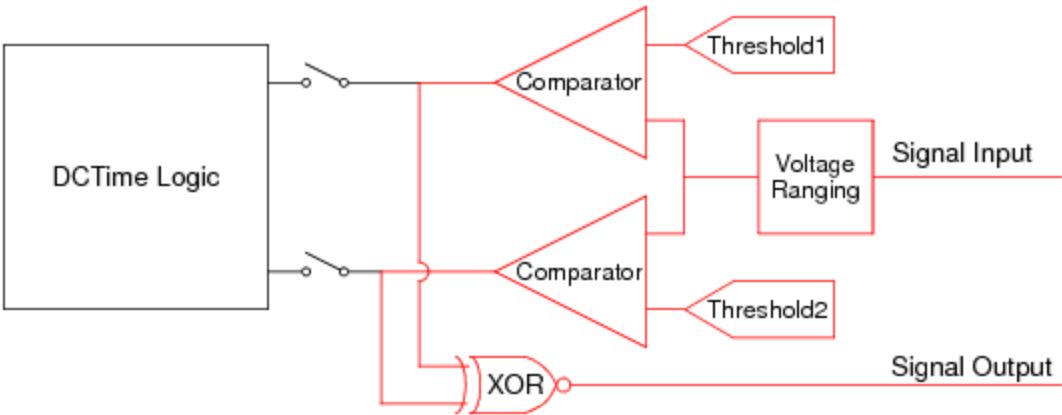
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DCTime Theory of Operation : Asynchronous Trigger : DIB Triggering

DIB Triggering

The DCTime front end can output a signal to the DIB for triggering of another instrument, DIB circuitry or a DUT pin. The comparator outputs are routed through an XOR gate, that is then placed on the DIB through the DCTime output. The DCTime output is DIB access on the DC30 and DC75. The DUT signal connects to the DCTime front end where the threshold level and voltage range can be programmed. When the threshold level is crossed a signal is output from the comparator to the XOR output. The figure shows the DIB trigger line connections:



DIB Asynchronous Trigger Lines

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DCTime Theory of Operation : DCTime Alarms

DCTime Alarms

An alarm indicates the instrument is not operating as programmed. This indicates that a condition exists that could potentially invalidate a measurement. DCTime alarms are:

- Capture
- CaptureNotTriggered
- DataLost
- DGS
- MeasurementIncomplete
- Timeout

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DCTime Theory of Operation : DCTime Alarms : Capture

Capture

An alarm occurred during a capture and the sample latched an alarm. Any of the DCTime alarms could have occurred while the capture was running.

Alarm Message

The following is the datalog alarm message:
{-1} : alarm [PinName, Strobe, SiteNum] TMU:0014 : Instrument alarmed during capture.
where:

PinName	Name of the pin the alarm occurred on.
SiteNum	Site number the alarm occurred on.

Common Causes

- An alarm condition occurred during the capture and may cause a measurement to be invalid.
- Based on the alarm, the instrument may be in a different condition than expected.

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DCTime Theory of Operation : [DCTime Alarms](#) : CaptureNotTriggered

CaptureNotTriggered

The expected event did not occur.

Alarm Message

The following is the datalog alarm message:

{-1} : alarm [PinName, Strobe, SiteNum] TMU:0014 : Capture not triggered or stamper event set to none when triggered.

where:

PinName	Name of the pin the alarm occurred on.
SiteNum	Site number the alarm occurred on.

Common Causes

- Hold off count not reached
- A capture is programmed with hold off enabled to a count of n, but the signal only makes n-1 transitions so no trigger occurs.

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DCTime Theory of Operation : DCTime Alarms : DataLost

DataLost

The capture overwrote the previous existing data due to a readback not occurring and no room was left in memory to retain all data.

Alarm Message

The following is the datalog alarm message:

{-1} : alarm [PinName, Strobe, SiteNum] Data lost. Memory was overwritten by subsequent captures.

where:

PinName	Name of the pin the alarm occurred on.
SiteNum	Site number the alarm occurred on.

Common Causes

- Samples overwritten before data is retrieved
- If 2000 samples is captured for a frequency measurement, before the data is retrieved 3000 samples for a duty cycle test is captured, the DCTime capture memory is exceeded and the first measurement is partly over written. When the samples from the frequency test are retrieved a Data lost alarm is issued.

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DCTime Theory of Operation : DCTime Alarms : DGS

DGS

Identifies a difference between the DGS and the DIB ground that exceeds instrument specifications.

Alarm Message

The following is the datalog alarm message:

{-1} : alarm [PinName, Strobe, SiteNum]TMU:0009 : DGS over voltage alarm in the hardware.

where:

PinName	Name of the pin the alarm occurred on.
SiteNum	Site number the alarm occurred on.

Common Causes

- | |
|----------------------------|
| <div>• Miswired DIB.</div> |
|----------------------------|

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DCTime Theory of Operation : [DCTime Alarms](#) : MeasurementIncomplete

MeasurementIncomplete

A measurement stopped before the required number of samples were captured. If a measurement is in progress and another Start microcode is executed, the current measurement stops and this alarm occurs.

Alarm Message

The following is the datalog alarm message:

{-1} : alarm [PinName, Strobe, SiteNum]TMU:0010 : Measurement incomplete alarm in the hardware.

where:

PinName	Name of the pin the alarm occurred on.
SiteNum	Site number the alarm occurred on.

Common Causes

- Can be caused by a trigger event occurring while a measurement is in progress.

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DCTime Theory of Operation : DCTime Alarms : Timeout

Timeout

The capture data was not available when the instrument was trying to retrieve it.

Alarm Message

The following is the datalog alarm message:

{-1} : alarm [PinName, Strobe, SiteNum] TMU:0013 : Timeout occurred while retrieving data.

where:

PinName	Name of the pin the alarm occurred on.
SiteNum	Site number the alarm occurred on.

Common Causes

- The data being retrieved took longer than the timeout setting.
- There is no data to retrieve.
- Two samples were requested but only one was taken. The hardware waits for the second sample for a max of 100ms or the programmed timeout.

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